



Retrieval scheduling in four-directional shuttle-based compact storage and retrieval systems with heterogeneous lifts

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ABSTRACT

As an emerging tier-to-tier shuttle-based system, the four-directional shuttle-based compact storage and retrieval system with heterogeneous lifts utilizes four-directional shuttles for in-tier bin transport; inter-tier movements are executed by two heterogeneous lift types, with shuttle-lifts transporting shuttles and bin-lifts transporting bins. Since the bin-lift is typically less expensive than the shuttle-lift, employing two types of lifts can reduce both investment and operational costs. The execution of retrieval requests in this system requires coordinated operation of three types of equipment, making the scheduling optimization problem both complex and challenging. This paper investigates the retrieval scheduling problem, which focuses on determining the task sequences of shuttles and lifts to minimize the makespan. We formulate a mixed integer programming model and propose an exact logic-based Benders decomposition algorithm. Additionally, we develop an effective two-stage heuristic to provide near-optimal solutions for large-scale instances. The two-stage heuristic consistently outperforms alternative scheduling strategies, delivering shorter makespan, lower average retrieval time, and reduced capital investment. Numerical experiments reveal that increasing the number of shuttles and bin-lifts, rather than shuttle-lifts, leads to a more significant reduction in makespan. In addition, bin-lifts demonstrate greater cost-effectiveness compared to shuttle-lifts; partial substitution of shuttle-lifts with bin-lifts yields substantial economic benefits, with the highest benefits observed when bin-lifts account for 20% of the total lift investment. Furthermore, positioning lifts at the midpoint along the picking tracks produces the minimum makespan. These findings provide valuable insights into equipment configuration for optimizing the performance of such systems.

1. Introduction

Automated storage and retrieval systems (AS/RSs) play a critical role in modern warehousing by reducing labor costs, enhancing reliability, and improving operational efficiency (Roodbergen & Vis, 2009). Among them, shuttle-based storage and retrieval systems (SBS/RSs) have gained popularity for their high throughput, with tier-captive SBS/RSs (c-SBS/RSs) being a widely studied configuration (Lerher, 2015; Tappia et al., 2016; Zou et al., 2016). However, the need for a dedicated shuttle per tier results in high equipment costs. To address this, tier-to-tier SBS/RSs (t-SBS/RSs) allow shuttles to move between tiers via lifts, improving utilization and reducing congestion (Jerman et al., 2021; Lerher et al., 2021). A further refinement introduces two types of lifts: shuttle-lifts for transporting shuttles and bin-lifts for handling bins, which reduces reliance on costly shuttle-lifts and improves overall system cost-efficiency (Arslan & Ekren, 2023). More

recently, four-directional SBS/RSs with two lift types (F-SBS/RSs-TL) have emerged as a compact, flexible solution. They are already adopted in pharmaceuticals, electronics, food and beverage, and footwear and apparel, with deployments reported in Enfon (2025) and Wayzim (2025). However, the intricate interactions among shuttles, shuttle-lifts, and bin-lifts make retrieval scheduling highly challenging; efficient scheduling strategies are therefore essential to fully realize the system's benefits.

Most prior work on retrieval scheduling in AS/RSs has focused on single-deep installations, where the retrieval scheduling primarily coordinates a single transport resource (for example, cranes or lifts) to satisfy requests while minimizing makespan (Carlo & Vis, 2012; Gharehgozli et al., 2017; Zhao et al., 2018). In contrast, research on multi-deep systems is relatively sparse; a few studies consider settings with two interacting resources (Dong et al., 2021; Yang et al.,

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2023), yet none address joint task allocation across three heterogeneous resources. F-SBS/RSs-TL rely on three distinct resources: (i) four-directional shuttles for horizontal bin movement within tiers, (ii) shuttle-lifts for inter-tier shuttle transport, and (iii) bin-lifts for vertical bin transport. The tight interdependence among these resources creates a multi-resource synchronization challenge that existing retrieval scheduling models cannot directly handle. With F-SBS/RSs-TL increasingly deployed in high-density warehouses, there is a pressing need for novel scheduling methods that jointly coordinate shuttles, shuttle-lifts, and bin-lifts to optimize system performance.

Fig. 1(a) presents the overall schematic of the F-SBS/RS-TL, a multi-tier storage system with three transport resources that serve distinct functions: (i) four-directional shuttles that move bins horizontally within a tier (Fig. 1(c)); (ii) bin-lifts, positioned on vertical tracks near the I/O points, that transfer bins vertically between storage tiers and input/output (I/O) points (Fig. 1(b)); and (iii) shuttle-lifts, located at the rear of the system, that reposition shuttles between tiers along vertical tracks (Fig. 1(d)). This study addresses the retrieval scheduling problem in this setting, seeking to minimize makespan by coordinating these three resources. In high-throughput warehousing environments, system makespan, defined as the total time required to complete a set of retrieval requests, is a critical performance metric. Decision-makers prioritize makespan minimization over other objectives (e.g., cost or distance minimization) because it directly affects system throughput, order fulfillment speed, and overall system responsiveness. The research question addressed in this paper is: Given a set of retrieval requests, how can we jointly determine the request sequence and integrated schedules for shuttles, shuttle-lifts, and bin-lifts to minimize system makespan? The core of this question is coordinating three tightly interdependent resources. Addressing it requires an integrated scheduling framework that jointly optimizes task assignment, sequencing, and cross-resource synchronization for shuttles, shuttle-lifts, and bin-lifts.

We develop a novel mixed-integer programming (MIP) model to precisely formulate the retrieval scheduling problem in an F-SBS/RS-TL. This model comprehensively captures the intricate interdependencies among the three types of storage/retrieval (S/R) equipment, enabling systematic optimization of retrieval scheduling decisions. Building on this mathematical formulation, we propose a tailored logic-based Benders decomposition (LBBD) algorithm. The Benders master problem optimizes the execution order of retrieval requests and the task sequencing of shuttles, and the Benders subproblem further refines the task sequences for the two types of lifts. To handle large-scale retrieval scheduling instances, we design a two-stage heuristic, referred to as ALNS&TS, which integrates adaptive large neighborhood search (ALNS) and Tabu search (TS). ALNS enhances solution exploration by dynamically adjusting search neighborhoods, while TS strengthens exploitation by refining promising solutions, thereby striking a balance between computational efficiency and solution quality. Together, these methodologies provide both exact and heuristic solutions that significantly improve the efficiency and scalability of retrieval scheduling in the F-SBS/RS-TL. By bridging the research gap in scheduling optimization for F-SBS/RS-TL, this study aims to enhance system performance and provide practical insights for warehouse automation. Our findings offer valuable guidance for optimizing retrieval operations, ultimately contributing to the broader development of intelligent warehousing and logistics automation.

Specifically, our contributions are summarized as follows:

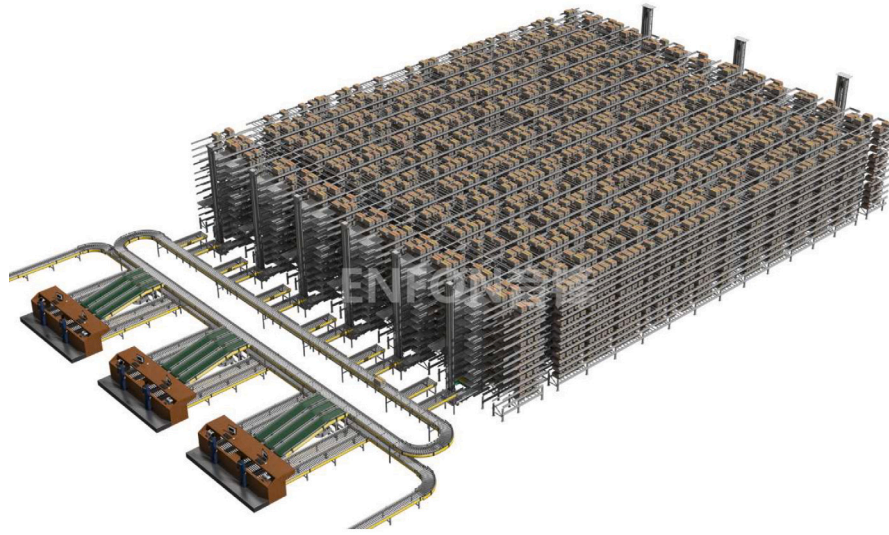
- We are among the first to formulate the retrieval scheduling problem that jointly decides request sequencing and the coordinated schedules of shuttles, shuttle-lifts, and bin-lifts in the F-SBS/RS-TL. The formulation explicitly captures interlocking precedence relations and synchronization constraints across the three resources, providing a unified optimization view of the system.
- We develop an exact LBBD algorithm with proven optimality cuts, which establishes a certified baseline and clarifies structural optimality properties of the scheduling problem. Complementing this, we design a two-stage ALNS&TS heuristic that rapidly produces high-quality schedules for large instances, making the approach directly usable in time-sensitive operational settings.
- Enabled by our structured optimization framework (modeling and decomposition), we conduct extensive, scalable numerical experiments; the resulting evidence provides multi-dimensional, actionable guidance for S/R equipment configuration and layout design.

2. Literature review

The retrieval scheduling problem in automated warehousing systems mainly involves two categories: retrieval performed by single S/R equipment and retrieval performed by cooperative work of multiple S/R equipment.

The retrieval scheduling problem of single S/R equipment typically occurs in AS/RSs, where one or multiple cranes are responsible for retrieving and storing bins. For a general review of the scheduling of single S/R equipment in AS/RSs, we refer to Boysen and Stephan (2016). The objective of scheduling the cranes is predominantly minimizing the total travel time. Han et al. (1987) first formulate the basic S/R equipment scheduling problem in AS/RSs, and show that the scheduling problem is analogous to the traveling salesman problem (TSP). Vis and Roodbergen (2009) sequence storage and retrieval requests of containers and reformulate the problem as an asymmetric Steiner TSP. To minimize total cost of the greenhouse gas efficiency, Nia et al. (2017) dynamically sequences the storage and retrieval requests in unit-load multiple-rack AS/RSs. Considering two-depot in a AS/RS, Gharehgozli et al. (2017) investigate the storage and retrieval sequencing problem to minimize the total travel time of S/R equipment, and design polynomial-time algorithms to solve the problem. With the objective of minimization of both total travel time and total tardiness, Man et al. (2021) focus on the S/R equipment sequencing problem in a two-depot AS/RS. Polten and Emde (2022) reformulate the multi-shuttle crane scheduling problem in AS/RSs as a capacitated vehicle routing problem, and propose an exact algorithm to solve it. Several studies have delved into various warehousing systems that utilize a single type of S/R equipment. In particular, Boysen et al. (2017) formalize a retrieval sequencing problem in mobile rack systems, and develop exact and heuristic algorithms to solve the model.

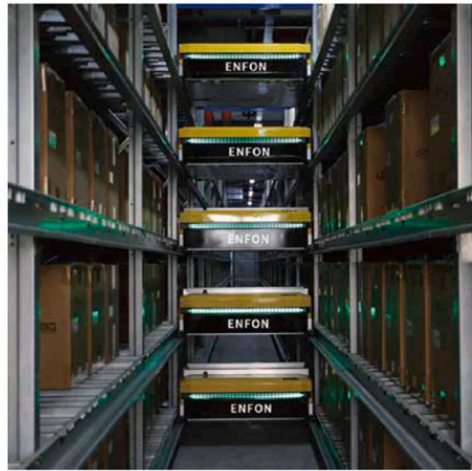
Several literature investigates the cooperative retrieval problem of multiple S/R equipment in autonomous vehicle storage and retrieval systems (AVS/RSs) and SBS/RSs, where the vehicles/shuttles and lifts independently conduct the horizontal and vertical movements respectively. Concerning the assignment of the vehicles/shuttles to the tiers, two types of system configuration are applied in practice: system with tier-to-tier vehicles/shuttles and system with tier-captive vehicles/shuttles. Wang et al. (2020) propose a mixed integer linear programming (MILP) model to describe the retrieval scheduling problem in a c-SBS/RS, and apply three heuristics to solve the model. Dong et al. (2021) investigate the retrieval scheduling problem in crane-based storage systems with shuttles, and develop four heuristics to address the scheduling problem. Yu et al. (2023) formulate the retrieval sequencing problem in c-AVS/RSs and propose an exact dynamic programming algorithm with a computational complexity of $O(N * 2^N)$, where N denotes the number of retrieval requests. (Yang et al., 2023) focus on optimizing the shuttle transfer sequence and the retrieval request sequence in a deep-lane SBS/RS with forklifts, and propose a two-stage heuristic to solve the mathematical model. Kosanić et al. (2023) develop travel time models for multi-deep tier-captive SBS/RSs, taking into account both single-command and dual-command operations. Marolt et al. (2023) further explore the combination of depth-first storage and relocation strategies, providing an analytical model for



(a) System Schematic



(b) Bin-lift



(c) Four-directional Shuttle



(d) Shuttle-lift

Fig. 1. A Four-directional SBS/RS with Two Types of Lifts (Enfon, 2025).

optimizing multi-deep AVS/RSs. Ekren et al. (2024) introduces a novel AVS/RS with movable lifts and compares it with an equivalent tier-captive SBS/RS, indicating that the AVS/RS with movable lifts may hold potential for the design of high-tier warehouse systems.

Applying two lifts in SBS/RSs is a relatively advanced technology, but a limited number of studies have investigated the retrieval scheduling problem in such SBS/RSs. Carlo and Vis (2012) propose a look-ahead strategy to determine the assignment of retrieval requests to lifts and the sequence of retrieval requests in a SBS/RS with two lifts. Zhao et al. (2018) further extend the research of Carlo and Vis (2012) by taking into account the acceleration and deceleration of the lifts, and apply a genetic algorithm to optimize the sequence of storage and retrieval requests, the scheduling of lifts, as well as the collision avoidance. Using deep Q-Learning (DQL) method, Arslan and Ekren (2023) address the task selection and sequencing problem in t-SBS/RSs. Ekren and Arslan (2024) implement a reinforcement learning based algorithm to sequence the storage and retrieval requests in SBS/RSs, and conclude that the proposed method outperforms the first-in-first-out rule. Considering both the lift-shuttle interaction and lift-lift interaction, Chen et al. (2023) formulate the retrieval scheduling problem in the SBS/RS with two lifts as an MIP model and develop a heuristic method to solve it.

To the best of our knowledge, this is the first study to model the retrieval scheduling problem in the F-SBS/RS-TL. Additionally, we develop both an exact algorithm and an efficient heuristic to solve the proposed model. Due to the inclusion of three types of S/R equipment with distinct functions, the problem formulation and algorithm design for the F-SBS/RS-TL are significantly more complex than those for systems with only two types of equipment, which have been extensively studied in prior research. We identify only one article, Zhen et al. (2023), that investigates the storage request scheduling problem in a system with three types of S/R equipment, and it only proposes a heuristic algorithm to solve the problem. Table 1 summarizes the related literature on retrieval scheduling problems in automated warehousing systems.

3. Problem formulation

In this section, we formally describe the retrieval scheduling problem in the F-SBS/RS-TL (Section 3.1). Based on this, we develop a mathematical model with the objective of minimizing the makespan (Section 3.2).

Table 1
Positioning of this research.

Citation	System	Type of S/R equipment	Retrieval sequencing	Equipment scheduling	Algorithm
Vis and Roodbergen (2009)	Container	One	✓		Exact
Gharehgozli et al. (2017)	AS/RS	One	✓		Exact
Boysen et al. (2017)	Mobile rack	One	✓		Exact & Heuristic
Zhao et al. (2018)	t-SBS/RS	Two	✓	✓	Heuristic
Wang et al. (2020)	c-SBS/RS	Two	✓	✓	Heuristic
Yu et al. (2023)	AVS/RS	Two	✓	✓	Exact & Heuristic
Yang et al. (2023)	SBS/RS	Two	✓	✓	Heuristic
Arslan and Ekren (2023)	t-SBS/RS	Three	✓		DQL
Zhen et al. (2023)	SBS/RS	Three		✓	Heuristic
This paper	F-SBS/RS-TL	Three	✓	✓	Exact & Heuristic

3.1. Problem description

This paper investigates the retrieval scheduling problem in the F-SBS/RS-TL, focusing on how to optimize the execution order of retrieval requests and the task sequence of S/R equipment to minimize the makespan. Makespan, defined as the maximum time required to complete all retrieval requests, directly reflects the total operation cycle of the system. Reducing makespan typically means completing all requests in the shortest possible time, thus enhancing overall operational efficiency. While reducing total travel time or total travel distance can also improve efficiency, focusing solely on these objectives in a complex system with multiple requests and equipment may result in an imbalanced distribution of tasks, causing certain equipment to be overloaded. This can undermine overall system performance and stability. In contrast, optimizing makespan accounts for the total time required to complete all tasks, providing a more holistic measure of system performance that reflects the effective scheduling of requests and the balanced allocation of resources.

Fig. 2 illustrates the front and side views of a double-deep F-SBS/RS-TL. In the diagram, blue squares represent storage bins, orange components represent shuttles, and rectangles with varying shades represent the two types of lifts. Additionally, the operating tracks of the shuttles, shuttle-lifts, and bin-lifts are clearly marked. Unlike traditional automated warehousing systems, the investigated system enables the synchronous operation of shuttles and two distinct types of lifts. To fully leverage the advantages of this configuration, it is essential to coordinate the scheduling of shuttles and lifts during bin retrieval tasks. System throughput is influenced not only by the number of shuttles and lifts but also by the effectiveness of their scheduling. A well-designed scheduling strategy can minimize the makespan, thereby reducing both investment and operational costs while maximizing system efficiency.

This paper focuses exclusively on retrieval requests due to their time-sensitive nature and the need for rapid responsiveness (Boysen et al., 2019; Yang et al., 2023). The retrieval scheduling problem seeks to minimize the time required to complete all retrieval requests. These requests are processed collaboratively by shuttles and two types of lifts: shuttles handle horizontal transportation of bins, bin-lifts perform vertical transportation, and shuttle-lifts facilitate tier changes for shuttles. Bins enter and leave the storage area via I/O points, uniformly located at the lower bottom of the system. Each retrieval request is assigned both a shuttle and a bin-lift (if the request is not located on the first tier), while each shuttle-transfer task is allocated to a shuttle-lift. The retrieval process is divided into three key stages: the horizontal-retrieval process, where shuttles transport bins to the bin-lift buffer; the vertical-retrieval process, where bin-lifts deliver bins to the I/O points; and the shuttle-transfer process, where shuttle-lifts move shuttles between tiers. This interdependent coordination among shuttles and the two types of lifts is crucial for the efficient execution of retrieval requests.

To streamline the modeling process, the execution of each retrieval request is divided into two distinct tasks:

(i) *Task 1*: An idle shuttle is assigned to the retrieval request. The shuttle travels from its current dwelling location to the retrieval

location. If the shuttle is not on the same tier as the retrieval location, a shuttle-lift is utilized to transfer the shuttle to the target tier.

(ii) *Task 2*: An idle bin-lift is assigned to the retrieval bin. The shuttle then transports the bin to the bin-lift buffer, after which the bin-lift transfers the bin from the storage tier to the I/O point.

To provide a detailed illustration of the retrieval scheduling problem, we design a specific instance to describe the execution process of retrieval requests, as depicted in Fig. 3. The retrieval scheduling problem involves three key scheduling decisions, corresponding to the horizontal-retrieval process, the shuttle-transfer process, and the vertical-retrieval process. In this instance, we consider five retrieval requests, three shuttles, two bin-lifts, and two shuttle-lifts. The objective is to schedule the shuttles and lifts in a way that minimizes the total completion time for all five retrieval requests. When a set of customer orders arrives, the system generates retrieval requests to fetch the required bins (i.e., *B1–B5* in Fig. 3). For each retrieval request, the system assigns a shuttle to transport the bin from its retrieval location to the bin-lift buffer. Subsequently, a bin-lift is utilized to transfer the bin from the buffer to the I/O point. This whole process is referred to as the retrieval execution process. Since shuttles are unable to move across tiers independently, the horizontal-retrieval task requires the assistance of shuttle-lifts. Taking retrieval request *B1* as an example, the shuttle *S1* is assigned to handle this request, with the shuttle-lift *SL1* facilitating *S1*'s tier transfer. Once *S1* transports bin *B1* to the buffer of bin-lift *BL1*, the bin-lift *BL1* completes the process by transferring bin *B1* to the I/O point. Through effective coordination of shuttles and lifts, the goal is to minimize the maximum completion time of all retrieval requests.

The assumptions for this study are as follows:

- Each lane is designated for storing a single type of item, as described in Zaerpour et al. (2015) and Zou et al. (2018). The reshuffling operation, which typically arises when multiple item types are stored in the same lane, is not considered.
- Both shuttles and lifts operate based on the Point of Service Completion (POSC) policy. Specifically, shuttles begin and end their operations at the bin-lift buffer, bin-lifts start and finish at the I/O point, and shuttle-lifts operate according to the tier of the retrieval location.
- Shuttles and lifts maintain constant horizontal and vertical movement speeds, respectively, as assumed in De Koster et al. (2008).

3.2. Model formulation

Table 2 presents the notation used in the model, including parameters, sets, and variables. To facilitate understanding of the execution stages of a retrieval request, Fig. 4 provides a Gantt-style timeline schematic that illustrates the coordination among the three types of equipment, with key timestamps marked for clarity. Specifically, at time t_0 , the shuttle initiates its movement from the current storage location to the shuttle-lift buffer. At the same time, the shuttle-lift, initially occupied, proceeds to the same buffer once it becomes available

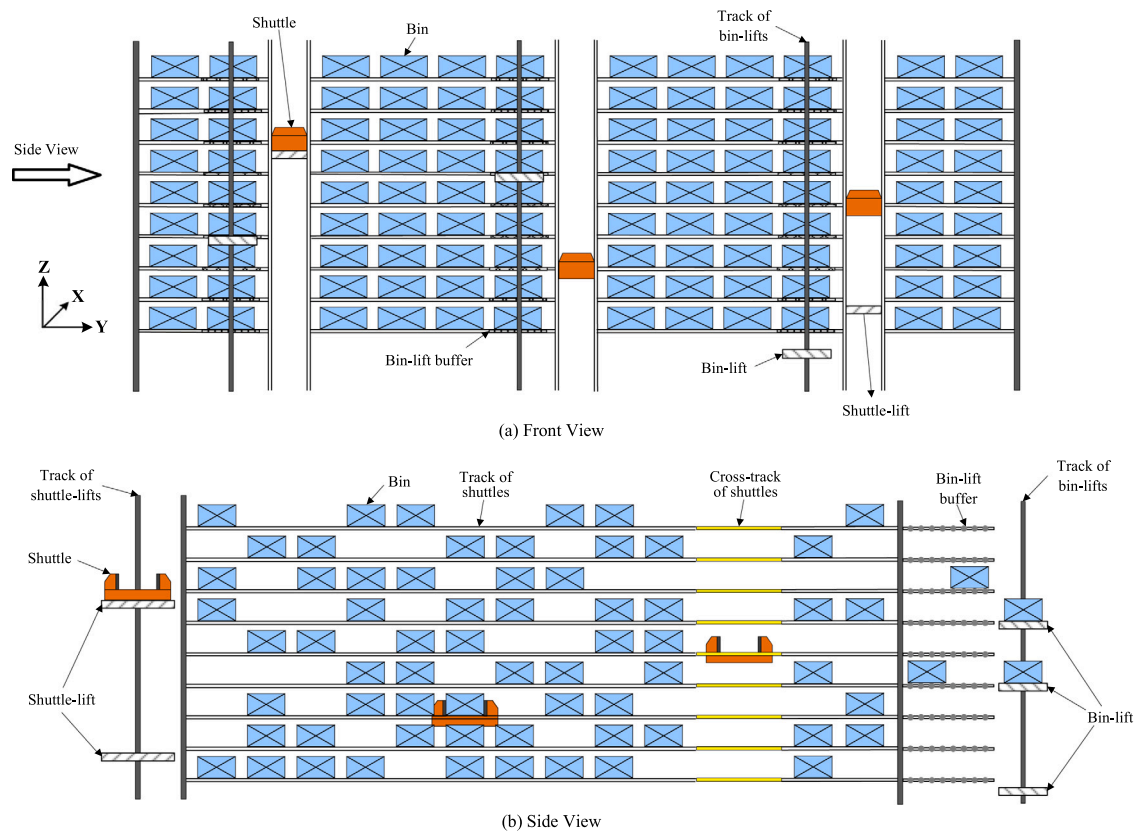


Fig. 2. Front and Side Views of the F-SBS/RS-TL.

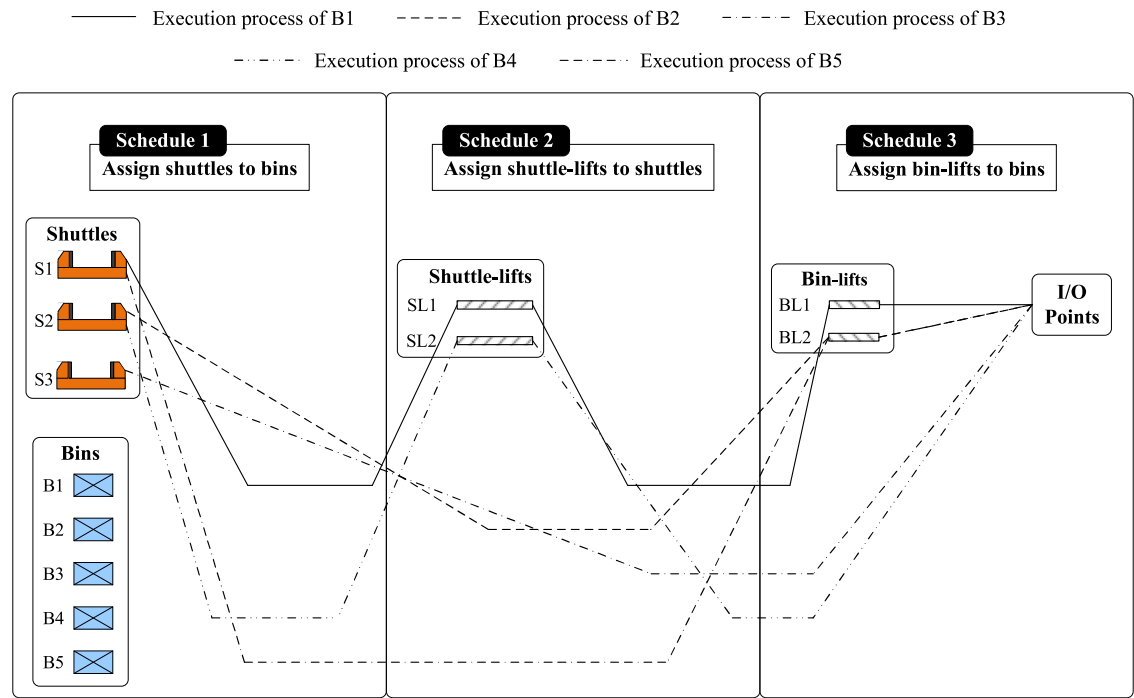


Fig. 3. Execution Process of Retrieval Requests.

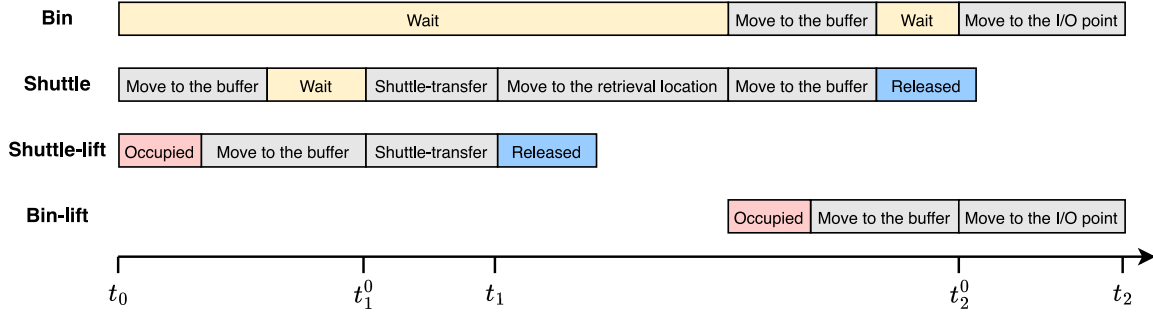


Fig. 4. Gantt Chart Illustrating the Retrieval Process and Equipment Coordination.

Table 2

Notation.

Notation	Description
Parameters	
$N_R, N_S, N_{L_S}, N_{L_B}$	Total number of retrieval requests, shuttles, shuttle-lifts, and bin-lifts
H, W, L	Warehouse height, width, and length (by number of bins)
h_e, w_i, l_e	Bin height, width, and length
v_S, v_{L_S}, v_{L_B}	The moving speeds of shuttles, shuttle-lifts, and bin-lifts
t_{lu}	The loading and unloading time of shuttles and lifts
tc_{ji}	Parameter taking the value 1 if request i is not on the same tier as request j
$xi(p), yi(p), zi(p), yt(p)$	The x-coordinate, y-coordinate, z-coordinate of location p , and the y-coordinate of the picking track to which location p belongs
yl_i^S, yl_i^B	The y-coordinates of shuttle-lift i and bin-lift i
rl_i	The retrieval location of request i
Sets	
$\mathcal{R}$	Set of all requests, $\mathcal{R} = \{1, 2, \dots, N_R\}$
$\mathcal{S}$	Set of all shuttles, $\mathcal{S} = \{1, 2, \dots, N_S\}$
$\mathcal{L}_S$	Set of all shuttle-lifts, $\mathcal{L}_S = \{1, 2, \dots, N_{L_S}\}$
$\mathcal{L}_B$	Set of all bin-lifts, $\mathcal{L}_B = \{1, 2, \dots, N_{L_B}\}$
$\mathcal{K}$	Set of all tasks in a request, $\mathcal{K} = \{1, 2\}$
Variables	
T	The time taken to complete all retrieval requests
t_{i1}, t_{i1}^0	The arrival time of the shuttle at the location of request i , and the start time of the corresponding shuttle-transfer task
t_{i2}, t_{i2}^0	The completion time of request i , and the start time of the corresponding vertical-retrieval task
x_{jis}	Binary variable taking the value 1 if request i is processed immediately after request j by shuttle s
$y_{jil_S}^S$	Binary variable taking the value 1 if request i is processed immediately after request j by shuttle-lift i
$y_{jil_B}^B$	Binary variable taking the value 1 if request i is processed immediately after request j by bin-lift i
yl_i^S	Auxiliary variable representing the y-coordinate of the shuttle-lift assigned to request i
yl_i^B	Auxiliary variable representing the y-coordinate of the bin-lift assigned to request i
z_{jis}^S	Auxiliary variable representing the product of variables x_{jis} and yl_i^S
z_{jis}^B	Auxiliary variable representing the product of variables x_{jis} and yl_i^B

to retrieve the shuttle. At time t_{i1}^0 , the shuttle-lift transports the shuttle to the designated tier and releases it. The shuttle then independently travels to the target bin location, arriving at time t_{i1} . After retrieving the bin, the shuttle carries it to the bin-lift buffer, where it waits until the bin-lift arrives at time t_{i2}^0 . The bin-lift subsequently delivers the bin to the I/O point at time t_{i2} , thereby completing the retrieval task through coordinated operations of the three types of equipment.

The objective function (1) aims to minimize the makespan, which represents the total time required to complete all retrieval requests. The constraints (2) specify the calculation of the makespan T . Alternatively,

other objective functions can also be considered depending on the system's operational priorities. For instance, minimizing the total travel distance, balancing workload among retrieval equipment, or minimizing energy consumption could be relevant in different application contexts.

$$\text{Min } T \quad (1)$$

$$\text{s.t. } T \geq t_{i2}, \quad \forall i \in \mathcal{R}, \quad (2)$$

The start time of request i is equal to the end time of request j , provided request j is processed immediately before request i by shuttle s (constraints (3)).

$$t_{i0} \geq t_{j2} + \left(\sum_{s \in \mathcal{S}} x_{jis} - 1 \right) M, \quad \forall i \in \mathcal{R}, j \in \mathcal{R} \cup \{0\}, \quad (3)$$

Constraints (4)~(5) are designed to enforce temporal consistency in the execution of shuttle-transfer tasks. Specifically, they ensure that the start time of a shuttle-transfer task corresponds to the later of two arrival times, namely the arrival time of the shuttle and that of the shuttle-lift at the shuttle-lift buffer. This condition reflects the operational requirement that the task cannot begin until both resources are present at the buffer. During model construction, constraints (4) are divided into four groups to address the presence of two absolute value terms, $|yl_i^S - yl_j^B|$ and $|yt(rl_i) - yl_j^B|$. Each group corresponds to a distinct combination of the sign conditions associated with these absolute value expressions, enabling a linear representation of the underlying piecewise structure. If the shuttle-transfer task is not required, t_{i1}^0 represents the time when the shuttle arrives at the retrieval location.

$$t_{i1}^0 \geq t_{i0} + \sum_{s \in \mathcal{S}} \sum_{j \in \mathcal{R} \cup \{0\}} x_{jis} (tc_{ji}((W+1)wi/v_S + |yl_i^S - yl_j^B| le/v_S) + (1 - tc_{ji})(xi(rl_i)wi/v_S + |yt(rl_i) - yl_j^B| le/v_S)), \quad \forall i \in \mathcal{R}, \quad (4)$$

$$t_{i1}^0 \geq t_{i1}^0 + \sum_{s \in \mathcal{S}} \sum_{j_1 \in \mathcal{R} \cup \{0\}} x_{j_1is} (|zi(rl_{j_1}) - zi(rl_j)| h_e/v_{L_S} + 2t_{lu}) + \left(\sum_{l_S \in \mathcal{L}_S} y_{jil_S}^S - 1 \right) M + |zi(rl_j) - zi(rl_i)| h_e/v_{L_S}, \quad \forall i \in \mathcal{R}, j \in \mathcal{R} \cup \{0\}, \quad (5)$$

After decomposing the absolute value terms, nonlinear expressions involving the product of variables, specifically $x_{jis} \cdot yl_i^S$ and $x_{jis} \cdot yl_j^B$, inevitably arise. Since x_{jis} is a binary variable, and both yl_i^S and yl_j^B are integer variables, auxiliary variables z_{jis}^S and z_{jis}^B are introduced to linearize these terms. The linearization procedure is detailed in constraints (6)~(11). Accordingly, constraints (4) can be reformulated into their linear equivalents, as shown in constraints (12).

$$z_{jis}^S \geq yl_i^S - M \cdot (1 - x_{jis}), \quad \forall j \in \mathcal{R} \cup \{0\}, i \in \mathcal{R}, s \in \mathcal{S}, \quad (6)$$

$$z_{jis}^S \leq M \cdot x_{jis}, \quad \forall j \in \mathcal{R} \cup \{0\}, i \in \mathcal{R}, s \in \mathcal{S}, \quad (7)$$

$$z_{jis}^S \leq yl_i^S, \quad \forall j \in \mathcal{R} \cup \{0\}, i \in \mathcal{R}, s \in \mathcal{S}, \quad (8)$$

$$z_{jis}^B \geq yl_j^B - M \cdot (1 - x_{jis}), \quad \forall j \in \mathcal{R} \cup \{0\}, i \in \mathcal{R}, s \in \mathcal{S}, \quad (9)$$

$$z_{jis}^B \leq M \cdot x_{jis}, \quad \forall j \in \mathcal{R} \cup \{0\}, i \in \mathcal{R}, s \in \mathcal{S}, \quad (10)$$

$$z_{jis}^B \leq y_{lj}^B, \quad \forall j \in \mathcal{R} \cup \{0\}, i \in \mathcal{R}, s \in \mathcal{S}, \quad (11)$$

$$\begin{aligned} t_{i1}^0 &\geq t_{i0} + \sum_{s \in \mathcal{S}} \sum_{j \in \mathcal{R} \cup \{0\}} tc_{ji}(x_{jis}(W+1)wi/v_S + |z_{jis}^S - z_{jis}^B| le/v_S) \\ &\quad + (1 - tc_{ji})(x_{jis} \cdot xi(rl_i)wi/v_S + |x_{jis} \cdot yi(rl_i) - z_{jis}^B| le/v_S), \\ &\quad \forall i \in \mathcal{R}, \end{aligned} \quad (12)$$

Similarly, constraints (13)~(14) ensure that the start time of a vertical-retrieval task equals the later of the shuttle's and the bin-lift's arrival times at the bin-lift buffer.

$$\begin{aligned} t_{i2}^0 &\geq t_{i1} + 2 |yi(rl_i) - yi(rl_i)| le/v_S + 2 \cdot t_{lu} + xi(rl_i)wi/v_S \\ &\quad + |yi(rl_i) - y_{li}^B| le/v_S, \quad \forall i \in \mathcal{R}, zi(rl_i) \geq 2, \end{aligned} \quad (13)$$

$$t_{i2}^0 \geq t_{j2} + (\sum_{l_B \in \mathcal{L}_B} y_{jil_B}^B - 1)M + (zi(rl_i) - 1)he/v_{L_B}, \quad \forall i \in \mathcal{R}, j \in \mathcal{R} \cup \{0\}, \quad (14)$$

Constraints (15) define the arrival time of the shuttle at the retrieval location. If $tc_{ji} = 0$, then $t_{i1} = t_{i1}^0$; otherwise, t_{i1} is computed as the sum of three components: the start time of the preceding shuttle-transfer task, the duration of that transfer task, and the travel time required for the shuttle to move independently from the shuttle-lift buffer to the retrieval location.

$$\begin{aligned} t_{i1} &\geq t_{i1}^0 + \sum_{s \in \mathcal{S}} \sum_{j \in \mathcal{R} \cup \{0\}} tc_{ji} \cdot x_{jis} (|zi(rl_j) - zi(rl_i)| he/v_{L_S} + 2 \cdot t_{lu} \\ &\quad + |yi(rl_i) - y_{li}^S| le/v_S + (W + 1 - xi(rl_i))wi/v_S), \quad \forall i \in \mathcal{R}, \end{aligned} \quad (15)$$

Constraints (16)~(17) specify the completion time of retrieval request i , distinguishing between two cases based on the vertical position of the retrieval location. Constraints (16) correspond to the scenario in which the retrieval location is on the first tier, that is, the same level as the I/O point, where bin-lift involvement is not required. In this case, the completion time equals the shuttle's arrival time at the retrieval location plus the time needed to move the bin from the retrieval location to the I/O point. Constraints (17) address the case in which the retrieval location is not on the first tier. In this situation, the completion time is determined by the start time of the vertical-retrieval task, plus the time required for that lift operation.

$$t_{i2} \geq t_{i1} + 2 |yi(rl_i) - yi(rl_i)| le/v_S + 4 \cdot t_{lu} + xi(rl_i)wi/v_S, \quad \forall i \in \mathcal{R}, zi(rl_i) = 1, \quad (16)$$

$$t_{i2} \geq t_{i2}^0 + (zi(rl_i) - 1)he/v_{L_B} + 2t_{lu}, \quad \forall i \in \mathcal{R}, zi(rl_i) \geq 2, \quad (17)$$

Constraints (18) ensure that each retrieval request is exclusively assigned to one shuttle, forming the foundation for constructing shuttle-specific task sequences. Building upon this, constraints (19)~(21) are designed to maintain the logical continuity and feasibility of task execution for each shuttle. Specifically, they guarantee that the tasks assigned to a given shuttle are sequentially connected, respecting both temporal precedence and resource availability, thus preventing infeasible task transitions or fragmented schedules.

$$\sum_{s \in \mathcal{S}} \sum_{j \in \mathcal{R} \cup \{0\} \setminus \{i\}} x_{jis} = 1, \quad \forall i \in \mathcal{R}, \quad (18)$$

$$\sum_{i \in \mathcal{R}} x_{0is} = 1, \quad \forall s \in \mathcal{S}, \quad (19)$$

$$\sum_{i \in \mathcal{R}} x_{i,N_R+1,s} = 1, \quad \forall s \in \mathcal{S}, \quad (20)$$

$$\sum_{j \in \mathcal{R} \setminus \{i\}} (x_{jis} - x_{ijs}) + x_{0is} - x_{i,N_R+1,s} = 0, \quad \forall i \in \mathcal{R}, s \in \mathcal{S}, \quad (21)$$

Similar to the shuttle task sequencing, constraints (22) ensure that each shuttle-transfer task is exclusively assigned to one shuttle-lift. Constraints (23)~(25) further enforce sequential feasibility by maintaining the temporal and logical continuity of tasks executed by each

shuttle-lift.

$$\sum_{l_S \in \mathcal{L}_S} \sum_{j \in \mathcal{R} \cup \{0\} \setminus \{i\}} y_{jil_S}^S \geq \sum_{s \in \mathcal{S}} \sum_{j \in \mathcal{R} \cup \{0\}} x_{jis} |zi(rl_j) - zi(rl_i)| / H, \quad \forall i \in \mathcal{R}, \quad (22)$$

$$\sum_{i \in \mathcal{R}} y_{0il_S}^S = 1, \quad \forall l_S \in \mathcal{L}_S, \quad (23)$$

$$\sum_{i \in \mathcal{R}} y_{i,N_R+1,l_S}^S = 1, \quad \forall l_S \in \mathcal{L}_S, \quad (24)$$

$$\sum_{j \in \mathcal{R} \setminus \{i\}} (y_{jil_S}^S - y_{ijl_S}^S) + y_{0il_S}^S - y_{i,N_R+1,l_S}^S = 0, \quad \forall i \in \mathcal{R}, l_S \in \mathcal{L}_S, \quad (25)$$

Finally, constraints (26) ensure that each retrieval bin located above the first tier is assigned to exactly one bin-lift. Furthermore, constraints (27)~(29) guarantee that the task sequence of bin-lifts is continuous.

$$\sum_{l_B \in \mathcal{L}_B} \sum_{j \in \mathcal{R} \cup \{0\} \setminus \{i\}} y_{jil_B}^B \geq (zi(rl_i) - 1)/H, \quad \forall i \in \mathcal{R}, \quad (26)$$

$$\sum_{i \in \mathcal{R}} y_{0il_B}^B = 1, \quad \forall l_B \in \mathcal{L}_B, \quad (27)$$

$$\sum_{i \in \mathcal{R}} y_{i,N_R+1,l_B}^B = 1, \quad \forall l_B \in \mathcal{L}_B, \quad (28)$$

$$\sum_{j \in \mathcal{R} \setminus \{i\}} (y_{jil_B}^B - y_{ijl_B}^B) + y_{0il_B}^B - y_{i,N_R+1,l_B}^B = 0, \quad \forall i \in \mathcal{R}, l_B \in \mathcal{L}_B, \quad (29)$$

Constraints (30)~(31) define the calculation of the auxiliary variables yl_i^S and yl_i^B , which represent the y-coordinates of the shuttle-lift and bin-lift assigned to request i , respectively. Constraints (32) specify the definition of the variables.

$$yl_i^S = \sum_{l_S \in \mathcal{L}_S} \sum_{j \in \mathcal{R} \cup \{0\}} y_{jil_S}^S \cdot yi^S(l_S), \quad \forall i \in \mathcal{R}, \quad (30)$$

$$yl_i^B = \sum_{l_B \in \mathcal{L}_B} \sum_{j \in \mathcal{R} \cup \{0\}} y_{jil_B}^B \cdot yi^B(l_B), \quad \forall i \in \mathcal{R}, \quad (31)$$

$$\begin{aligned} x_{jis} &\in \{0, 1\}, y_{jil_S}^S \in \{0, 1\}, y_{jil_B}^B \in \{0, 1\}, yl_i^S \in \{1, L\}, yl_i^B \in \{1, L\}, z_{jis}^B \in \{0, L\}, \\ z_{jis}^S &\in \{0, L\}, T \geq 0, t_{ik} \geq 0, t_{ik}^0 \geq 0, \quad \forall i \in \mathcal{R} \cup \{N_R + 1\}, j \in \mathcal{R} \cup \{0\}, \\ s &\in \mathcal{S}, l_S \in \mathcal{L}_S, l_B \in \mathcal{L}_B, k \in \mathcal{K}. \end{aligned} \quad (32)$$

Theorem 1. *The retrieval scheduling problem is NP-hard.*

Proof. To better understand the computational complexity of the retrieval scheduling problem (RSP), we analyze a simplified variant and establish its equivalence to a well-known NP-hard problem. Specifically, the general RSP is defined as $RSP(N_R, N_S, N_{L_S}, N_{L_B})$, where N_R denotes the total number of retrieval requests, N_S the number of shuttles, N_{L_S} the number of shuttle-lifts, and N_{L_B} the number of bin-lifts. For analytical tractability, we first focus on a system variant without shuttle-lifts. In this configuration, each retrieval request is located on the same tier as its assigned shuttle, thereby eliminating the requirement for shuttle-transfer operations. As a result, the problem reduces to $RSP(N_R, N_S, 0, N_{L_B})$, where N_S shuttles perform horizontal-retrieval tasks in the first stage, and N_{L_B} bin-lifts handle vertical-retrieval tasks in the second stage. Furthermore, when the system is equipped with only a single bin-lift, the problem further simplifies to $RSP(N_R, N_S, 0, 1)$. This special case can be equivalently transformed into a two-stage flexible flow-shop problem (TSFFSP), denoted as $TSFFSP(m, n)$, which consists of m jobs and two processing stages with n machines in the first stage and 1 machine in the second stage. The objective of the TSFFSP is to determine a job sequence minimizing the maximum completion time across all machines, which directly aligns with the objective of the RSP. Consequently, the instance $RSP(N_R, N_S, 0, 1)$ can be mapped to $TSFFSP(m, n)$ with $m = N_R$ and $n = N_S$. Given that the TSFFSP is well-known to be NP-hard (Hoogeveen et al., 1996), it follows that the RSP is also NP-hard.

4. Logic-based Benders decomposition method

Commercial solvers, such as Gurobi, are capable of providing optimal solutions for the retrieval scheduling problem; however, their computational time is typically long. Due to the complex combinatorial structure of the retrieval scheduling problem, the number of variables and constraints grows exponentially as the problem size increases, leading to significant computational challenges for commercial solvers. Therefore, to enhance solution efficiency, it is crucial to propose an algorithm that accelerates the generation of the optimal solution.

The retrieval scheduling model involves two distinct decision-making components: the task sequence for shuttles (denoted as $\mathbf{x}$) and the task sequences for shuttle-lifts ($\mathbf{y}^S$) and bin-lifts ($\mathbf{y}^B$). This structure allows the original model to be divided into two independent decision stages: the first stage focuses on solving the scheduling problem for shuttles, while the second stage determines the task sequences for shuttle-lifts and bin-lifts. The Benders decomposition approach, introduced by Benders (1962), is widely used in two-stage optimization problems. It separates the original problem into two independent components: the Benders master problem and the Benders subproblem.

In contrast to classical Benders decomposition, which requires the subproblem to be a linear program (LP) so that dual variables can be extracted and used to construct Benders cuts, the retrieval scheduling problem involves a combinatorial subproblem formulated as an MIP model. As such, no LP dual is available, rendering classical Benders decomposition inapplicable in this context. To overcome this limitation, we adopt the logic-based Benders decomposition (LBBD) framework (Hooker, 2000; Hooker & Ottosson, 2003), which generalizes the Benders paradigm by replacing LP dual with inference dual. LBBD allows the subproblem to be solved using MIP, and transmits logic-based (rather than dual-based) cuts to the master problem. This makes LBBD particularly well-suited for problems with complex combinatorial structures, such as the retrieval scheduling problem studied here, where strong inference cuts can be constructed based on problem-specific logic and solution structure (Roshanaei et al., 2020).

Using the LBBD approach, we formulate a Benders master problem to address the scheduling of shuttles (the first-stage problem) and a Benders subproblem to handle the scheduling of shuttle-lifts and bin-lifts, given a trial task sequence for the shuttles (the second-stage problem). Fig. 5 illustrates how the LBBD approach works. Specifically, if the master problem returns the same X^* in the next iteration, the cut ensures that the corresponding incremental objective value of the makespan must be not less than q^* . In doing so, the cut prevents the algorithm from re-selecting the same master decision without making progress on the lower bound. This mechanism effectively tightens the lower bound and guides the master problem toward exploring new and potentially better solutions.

4.1. Benders master problem

The Benders master problem, defined by Eqs. (33) ~ (41), addresses the first-stage problem, focusing on assigning shuttles to retrieval requests to derive an optimal task sequence for the shuttles. In this formulation, variables and constraints associated with the second-stage problem are excluded. Furthermore, a variable q is introduced to estimate the additional makespan contributed by the shuttle-lifts and bin-lifts.

$$\text{Min } Z = T + q \quad (33)$$

$$\text{s.t. Constraints (2) ~ (3), (18) ~ (21),} \quad (34)$$

$$t_{i1}^0 \geq t_{i0} + \sum_{s \in S} \sum_{j \in \mathcal{R} \cup \{0\}} x_{jis} (tc_{ji} (W + 1)wi/v_S + (1 - tc_{ji})xi(r_l)wi/v_S), \quad \forall i \in \mathcal{R}, \quad (35)$$

$$t_{i2}^0 \geq t_{i1} + 2 | yt(r_l) - yi(r_l) | le/v_S + 2 \cdot t_{lu} + xi(r_l)wi/v_S, \quad \forall i \in \mathcal{R}, \quad (36)$$

$$t_{i1} \geq t_{i1}^0 + \sum_{s \in S} \sum_{j \in \mathcal{R} \cup \{0\}} tc_{ji} \cdot x_{jis} (| zi(r_l) - zi(r_l) | he/v_{L_S} + 2t_{lu} + (W + 1 - xi(r_l))wi/v_S), \quad \forall i \in \mathcal{R}, \quad (37)$$

$$t_{i2} \geq t_{i2}^0 + (zi(r_l) - 1)he/v_{L_B} + 2t_{lu}, \quad \forall i \in \mathcal{R}, zi(r_l) \geq 2, \quad (38)$$

$$t_{i2} \geq t_{i1} + 2 | yt(r_l) - yi(r_l) | le/v_S + 2 \cdot t_{lu} + xi(r_l)wi/v_S, \quad \forall i \in \mathcal{R}, zi(r_l) = 1, \quad (39)$$

$$\text{LBBD optimality cut,} \quad (40)$$

$$x_{jis} \in \{0, 1\}, T \geq 0, q \geq 0, t_{ik} \geq 0, t_{ik}^0 \geq 0. \quad \forall i \in \mathcal{R} \cup \{N_R + 1\}, j \in \mathcal{R} \cup \{0\}, s \in S, k \in \mathcal{K}. \quad (41)$$

The objective (33) seeks to minimize the system makespan. Considering only the operation of shuttles, constraints (35)~(36) determine the start times for the shuttle-transfer and vertical-retrieval tasks, respectively. Constraints (37)~(39) define the shuttle's arrival time at the retrieval location and the completion time for retrieval request i , respectively. Constraint (40) represents the LBBD optimality cut, which is iteratively generated by the Benders subproblem. A detailed explanation of the LBBD optimality cut is provided in Section 4.3. Lastly, constraints (41) define the decision variables.

4.2. Benders subproblem

Using the trial values $\hat{\mathbf{x}}$ obtained from the Benders master problem, the retrieval scheduling problem is simplified into a Benders subproblem that focuses on scheduling shuttle-lifts and bin-lifts. The Benders subproblem aims to determine the optimal execution order for shuttle-transfer tasks and vertical-retrieval tasks. The detailed formulation of the Benders subproblem is presented below.

$$\text{Min } T \quad (42)$$

$$\text{s.t. Constraints (2), (13) ~ (15), (17), (23) ~ (31),} \quad (43)$$

$$t_{i0} \geq t_{j2} + (\sum_{s \in S} \hat{x}_{jis} - 1)M, \quad \forall i \in \mathcal{R}, j \in \mathcal{R} \cup \{0\}, \quad (44)$$

$$t_{i1}^0 \geq t_{i0} + \sum_{s \in S} \sum_{j \in \mathcal{R} \cup \{0\}} \hat{x}_{jis} (tc_{ji} (W + 1)wi/v_S + | yl_i^S - yl_j^B | le/v_S) + (1 - tc_{ji})(xi(r_l)wi/v_S + | yt(r_l) - yl_j^B | le/v_S), \quad \forall i \in \mathcal{R}, \quad (45)$$

$$t_{i1}^0 \geq t_{j1}^0 + \sum_{s \in S} \sum_{j_1 \in \mathcal{R} \cup \{0\}} \hat{x}_{j_1js} (| zi(r_{l_{j_1}}) - zi(r_l) | he/v_{L_S} + 2 \cdot t_{lu}) + (\sum_{l_S \in \mathcal{L}_S} y_{jil_S}^S - 1)M + | zi(r_l) - zi(r_l) | he/v_{L_S}, \quad \forall i \in \mathcal{R}, j \in \mathcal{R} \cup \{0\}, \quad (46)$$

$$t_{i1} \geq t_{i1}^0 + \sum_{s \in S} \sum_{j \in \mathcal{R} \cup \{0\}} tc_{ji} \cdot \hat{x}_{jis} (| zi(r_l) - zi(r_l) | he/v_{L_S} + 2 \cdot t_{lu} + | yt(r_l) - yl_i^S | le/v_S + (W + 1 - xi(r_l))wi/v_S), \quad \forall i \in \mathcal{R}, \quad (47)$$

$$\sum_{l_S \in \mathcal{L}_S} \sum_{j \in \mathcal{R} \cup \{0\} \setminus \{i\}} y_{jil_S}^S \geq \sum_{s \in S} \sum_{j \in \mathcal{R} \cup \{0\}} \hat{x}_{jis} | zi(r_l) - zi(r_l) | / H, \quad \forall i \in \mathcal{R}, \quad (48)$$

$$y_{jil_S}^S \in \{0, 1\}, y_{jil_B}^B \in \{0, 1\}, yl_i^S \in \{1, L\}, yl_i^B \in \{1, L\}, T \geq 0, t_{ik} \geq 0, t_{ik}^0 \geq 0, \quad \forall i \in \mathcal{R} \cup \{N_R + 1\}, j \in \mathcal{R} \cup \{0\}, l_S \in \mathcal{L}_S, l_B \in \mathcal{L}_B, k \in \mathcal{K}. \quad (49)$$

Considering both shuttle-transfer tasks and vertical-retrieval tasks, the objective (42) seeks to minimize the system makespan. Using the trial values $\hat{\mathbf{x}}$, the constraints (44) calculate the start time of retrieval request i . Constraints (45) and (46) update the start time for the shuttle-transfer task, while constraints (47) determine the arrival time of the shuttle at the retrieval location. Constraints (48) ensure that each

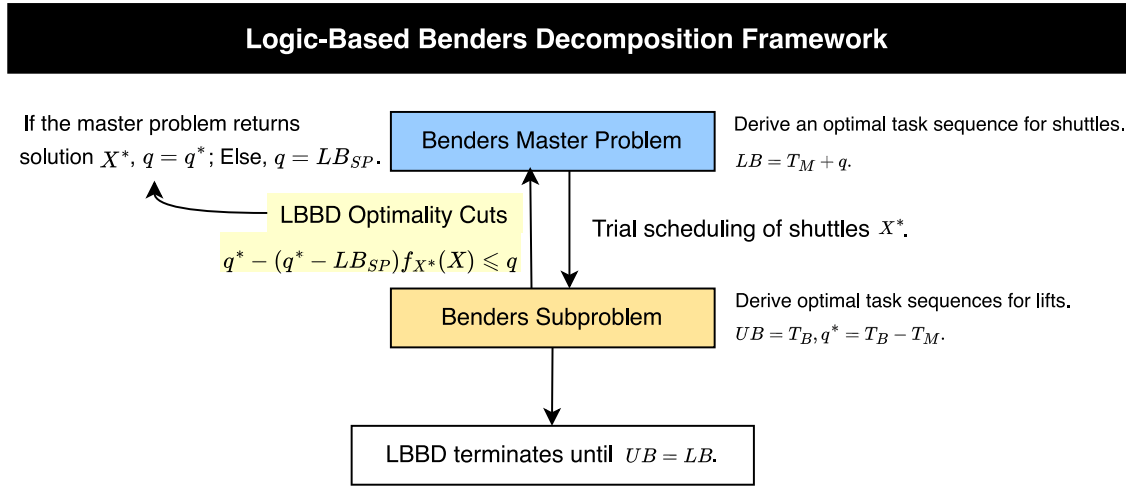


Fig. 5. Logic-based Benders Decomposition Framework.

shuttle-transfer task is assigned to a shuttle-lift. Lastly, constraints (49) define the variables.

4.3. LBBD optimality cuts

In the LBBD approach, two types of cuts are employed: feasibility cuts and optimality cuts, as in the classical Benders decomposition. When the Benders master problem leads to an infeasible subproblem, an LBBD feasibility cut is added to the master problem. LBBD optimality cuts, on the other hand, are used when the master problem either underestimates or overestimates the minimum or maximum objective value of the subproblem. For the retrieval scheduling problem, infeasible subproblems do not arise, as it is always possible to find a feasible scheduling scheme for the lifts (Benders subproblem) given any trial task sequence of shuttles (Benders master problem). Consequently, our focus is on deriving LBBD optimality cuts. Through the derivation process, we generate an optimality cut (Eq. (50)), which is iteratively added to the Benders master problem to refine the solution.

For simplicity, we denote the task sequence of shuttles as $X = \{x_{jis}, \forall i \in I, j \in R \cup \{0\}, s \in S\}$, the task sequence of shuttle-lifts as $Y^S = \{y_{jil}^S, \forall i \in I, j \in R \cup \{0\}, l \in L_S\}$, and the task sequence of bin-lifts as $Y^B = \{y_{jil}^B, \forall i \in I, j \in R \cup \{0\}, l \in L_B\}$. The optimality cut (Eq. (50)) provides a lower bound for q : the makespan increment corresponding to the task sequence of shuttles X^* is q^* . If the task sequence differs from X^* , the corresponding makespan increment is set to LB_{SP} . The value LB_{SP} is calculated as follows: First, assign the shuttle-lift closest to the shuttle's dwelling location for the shuttle-transfer task, and assign the bin-lift closest to the retrieval location for the vertical-retrieval task. Then, compute the makespan under this greedy scheme and set the difference between this makespan and the makespan obtained from the Benders master problem as LB_{SP} .

$$q^* - (q^* - LB_{SP})f_{X^*}(X) \leq q, \quad (50)$$

$$f_{X^*}(X) = \sum_{x \in \mathcal{X}_1} (1 - x) + \sum_{x \in \mathcal{X}_0} x, \quad (51)$$

where $\mathcal{X}_1^* = \{x | x_{jis}^* = 1, \forall i \in I, j \in R \cup \{0\}, s \in S\}$, and $\mathcal{X}_0^* = \{x | x_{jis}^* = 0, \forall i \in I, j \in R \cup \{0\}, s \in S\}$.

Theorem 2. The LBBD optimality cut (Eq. (50)) is valid, and satisfies the following two properties:

- $q^* - (q^* - LB_{SP})f_{X^*}(X^*) = q^*$
- Any feasible solution $(\bar{X}, \bar{Y}^S, \bar{Y}^B)$ of the problem that $\bar{X} \neq X^*$ satisfies $q^* - (q^* - LB_{SP})f_{X^*}(\bar{X}) \leq q$

Proof.

- The first property follows from the definition of the optimality cut (Eq. (50)) that $f_{X^*}(X^*) = 0$.
- For $\bar{X} \neq X^*$, $f_{X^*}(\bar{X}) \geq 1$, we have $q^* - (q^* - LB_{SP})f_{X^*}(\bar{X}) \leq q^* - (q^* - LB_{SP}) = LB_{SP} \leq q$.

The LBBD approach improves solving speed by decomposing the problem into two stages. However, while effective for small- to medium-scale problems, its computational time remains relatively long for larger instances, making it challenging to meet the scheduling requirements of real-world scenarios. Therefore, for large-scale warehouses, it becomes essential to develop an efficient heuristic algorithm that accelerates the solving process while maintaining solution quality.

5. Two-stage heuristic method

The retrieval scheduling problem under investigation is NP-hard, characterized by a combinatorial solution space that grows exponentially with the number of retrieval tasks and the interdependencies among different types of S/R equipment. In this context, we define small-scale instances as those comprising fewer than 10 retrieval requests, medium-scale instances as those involving approximately 10 to 20 requests, and large-scale instances as those exceeding 20 requests. Commercial solvers such as Gurobi are capable of obtaining optimal solutions for small-scale instances within half an hour. For medium-scale instances, the proposed LBBD algorithm is able to extend the solvable range by efficiently pruning the search space. However, for large-scale instances, both the commercial solver and the LBBD algorithm face considerable computational difficulties, which stem from the problem's inherent combinatorial explosion and the intricate coupling between task assignments and resource constraints. To address this challenge, we design a two-stage heuristic algorithm, referred to as ALNS&TS, which combines ALNS and TS to enhance computational efficiency for large-scale retrieval scheduling problems. The motivation for employing ALNS and TS lies in their proven effectiveness in solving complex scheduling and routing problems, particularly those involving joint optimization of multiple interacting components, such as vehicle routing problems and machine scheduling (Francois et al., 2019; Li & Gao, 2016; Soriano et al., 2023).

ALNS is particularly well-suited for this problem due to its ability to dynamically adjust operator weights, ensuring adaptability across different instance characteristics. To tailor the algorithm to the F-SBS/RS-TL, we introduce problem-specific operators within ALNS, including the workload-aware operator, which redistributes retrieval tasks to balance the workload across shuttles, and the bin-clustering operator, which

groups retrieval requests based on spatial proximity to improve batch retrieval efficiency and reduce unnecessary movements. Meanwhile, TS enhances local search by utilizing a Tabu table to prevent cycling and premature convergence to local optima. By integrating the exploration capability of ALNS with the exploitation strength of TS, along with problem-specific operators that capture key system dynamics, the ALNS&TS algorithm efficiently navigates the complex solution space, providing high-quality retrieval scheduling solutions for large-scale instances.

This section introduces the ALNS&TS heuristic for solving the retrieval scheduling problem in F-SBS/RS-TL systems. The overall framework of the heuristic is outlined in Algorithm 1. Specifically, Section 5.1 presents the ALNS method, which tackles the first-stage problem by determining the execution order of retrieval requests and scheduling shuttles. To enhance the adaptability of the algorithm, we replace the traditional roulette wheel strategy for operator selection with a learning-based mechanism grounded in Q-learning. In this framework, each pair of destroy and repair operators is treated as an action, and the algorithm maintains a Q-value table to estimate the expected reward (e.g., improvement in solution quality) for each action based on past experiences. During the search process, the Q-values are continuously updated according to a reward signal, and operator pairs are selected using an ϵ -greedy strategy that balances exploration and exploitation. This learning-based selection enables the algorithm to adapt to instance characteristics and gradually favor more effective operators, improving solution quality and robustness. Based on the shuttle task sequences obtained from the ALNS stage, the second-stage problem is addressed using the TS method, as detailed in Section 5.2. This design loosely couples combinatorial routing with continuous-time scheduling and adjusts timing only near handover points. The search space shrinks and convergence becomes more stable. First, relative to a *lifts-first, shuttles-second* order, we do not lock lift sequences before arrival times are known, which reduces rework and the risk of infeasibility. Then, compared with a *three-stage split*, we introduce no additional interfaces and avoid duplicated routing, so coordination overhead remains low and cross-module reconciliation is rarely needed. Last, unlike *fully simultaneous optimization*, we retain key structural information while capping neighborhood propagation, which delivers substantial runtime savings with little loss in solution quality.

Algorithm 1 ALNS&TS Procedure

Input: Retrieval request set R , shuttle set S , shuttle-lift set L_S , bin-lift set L_B .
Output: Final task sequences: shuttles X_{best} , shuttle-lifts $Y S_{best}$, bin-lifts $Y B_{best}$.
 Initialize X_{cur} by greedy assignment of R to S .
while max iterations not reached **and** time limit not exceeded **do**
 $X_{temp} \leftarrow$ Apply selected destroy and repair operators to X_{cur} .
 if X_{temp} is feasible **then**
 $X_{cur} \leftarrow X_{temp}$
 Generate $Y S_{temp}$ via TS (see Algorithm 2). Then, compute t_{i2} and generate $Y B_{temp}$ accordingly.
 if $\text{makespan}(X_{temp}) < \text{makespan}(X_{best})$ **then**
 $X_{best} \leftarrow X_{temp}$
 end if
 end if
 Update operator weights.
end while

5.1. ALNS method for generating first-stage solutions

The ALNS method, initially proposed by Ropke and Pisinger (2006a), is an extension of the large neighborhood search method (Shaw, 1997). The core idea of ALNS involves a two-phase process: first, the solution is “destroyed”, and then it is “repaired”. The adaptive aspect of ALNS

lies in dynamically adjusting the weights of the operators used in these phases. By incorporating multiple sets of destroy and repair operators, the method expands the solution space and iteratively improves the best solution. The neighborhood of a solution is defined as a set of solutions derived by applying a destroy operator to partially dismantle the current solution and then using a repair operator to reconstruct it. Below, we describe three problem-specific destroy operators designed for this approach.

- **Random removal:** This operator removes retrieval requests randomly from the current solution, enhancing solution diversity and preventing premature convergence to local optima. By introducing randomness in the search process, this strategy enables the exploration of diverse solution spaces, increasing the likelihood of identifying high-quality retrieval schedules.
- **Worst removal:** Inspired by Ropke and Pisinger (2006b), this operator aims to intensify the search by removing retrieval requests that contribute the most to the system makespan. Specifically, the makespan impact of each retrieval request is evaluated by computing the difference in makespan before and after its removal. The retrieval requests are then ranked in descending order based on their individual contributions to the makespan. The requests with the highest impact are iteratively removed until the predefined removal threshold is met. By eliminating the most time-consuming retrievals, this strategy creates room for improved task reassignment and scheduling optimization.
- **Workload-aware removal:** This operator strategically removes retrieval requests assigned to the most heavily utilized shuttles to balance workload distribution across the system. First, the total task execution time of each shuttle is computed as a workload measure. The shuttles are ranked in descending order based on workload, and the retrieval requests assigned to the most burdened shuttle are selected for removal. The number of removed requests is determined based on predefined parameters. This approach alleviates potential bottlenecks caused by excessive workload concentration, leading to more balanced and efficient retrieval scheduling.

In addition, four problem-specific repair operators (random insertion, greedy insertion, workload-aware insertion, and bin-clustering insertion) are used in the ALNS method.

- **Random insertion:** This operator reassigns the removed retrieval requests to shuttles in a random manner, promoting solution diversity and preventing overly deterministic search patterns. By introducing randomness into the insertion process, this strategy helps explore a broader solution space and enhances the robustness of the heuristic.
- **Greedy insertion:** This insertion heuristic follows a myopic yet efficient approach by reinserting removed retrieval requests into the task sequence that results in the smallest increase in makespan. For each removed request, all possible insertion positions are evaluated, and the one leading to the least system-wide delay is selected. This strategy prioritizes local improvements and enables fast convergence toward high-quality solutions.
- **Workload-aware insertion:** To improve workload balance across shuttles, this operator redistributes removed retrieval requests by prioritizing assignment to the shuttle with the smallest current workload. Specifically, the shuttle with the lowest task execution time is selected as the insertion target. Each retrieval request is then reassigned in a way that minimizes the overall increase in makespan. This workload-aware insertion strategy helps mitigate bottlenecks and ensures a more evenly distributed retrieval workload.
- **Bin-clustering insertion:** This operator reconstructs a feasible retrieval schedule by leveraging spatial proximity among bins

to optimize retrieval efficiency. Retrieval requests are grouped into clusters based on their spatial locations, reducing redundant movements and improving batch retrieval performance. Once clustered, retrieval tasks are assigned to shuttles in a manner that balances workload across tiers. Within each cluster, retrieval sequences are further refined to minimize travel distances and waiting times, enhancing system throughput and reducing unnecessary equipment idling.

5.2. TS method for generating second-stage solutions

The TS method, a meta-heuristic algorithm based on local search, is originally proposed by [Glover \(1986\)](#). To escape local optima, TS allows the acceptance of inferior solutions, meaning that the solution obtained at each iteration is not necessarily better than the original one. To prevent loops, TS maintains a Tabu table that records the most recently accepted operations, which are then prohibited in subsequent iterations. Only solutions not listed in the Tabu table are considered as candidates for the next iteration. As the iterations progress, the Tabu table is dynamically updated, releasing the earliest operations after a specified number of iterations. This mechanism ensures diversification of the search space while maintaining computational efficiency. TS has been successfully applied to a wide range of combinatorial optimization problems, offering near-optimal solutions within a short computational time frame. For a comprehensive description of the TS method and its various extensions, see [Gendreau and Potvin \(2010\)](#).

Algorithm 2 TS Procedure

Input: Retrieval request set R , task sequence of shuttles X , shuttle-lift set L_S , and Tabu length b .
Output: Final task sequence of shuttle-lifts YS_{best} .
 Randomly assign shuttle-lifts to the shuttle-transfer tasks, and obtain an initial task sequence of shuttle-lifts YS_{cur} . Initialize the Tabu list.
while max iterations not reached **and** time limit not exceeded **do**
 Generate the neighborhood of the current solution YS_{cur} and select a candidate solution YS_{temp} .
 if YS_{temp} is not forbidden in the Tabu table, or the system makespan of YS_{temp} is less than that of YS_{best} **then**
 $YS_{cur} \leftarrow YS_{temp}$.
 if the system makespan of YS_{temp} is less than that of YS_{best} **then**
 $YS_{best} \leftarrow YS_{temp}$.
 end if
 end if
 Update the Tabu table.
end while

6. Numerical experiments

The numerical experiments are implemented in Java, with Gurobi (version 9.5.1, [Gurobi Optimizer, 2025](#)) utilized to solve the mathematical models. All experiments are conducted on a workstation equipped with an Intel Xeon Silver 4210R CPU (2.40 GHz) and 64 GB of RAM.

6.1. Instance generation

To evaluate the effectiveness of the exact LBB algorithm and the two-stage heuristic, we generate nine sets of instances to analyze their computational time and solution quality across various scenarios. Parameters for each instance are randomly generated, referencing both existing literature and practical applications. [Table 3](#) details the number of retrieval requests (N_R), shuttles (N_S), shuttle-lifts (N_{L_S}), and bin-lifts (N_{L_B}) for each instance set. Additionally, [Table 4](#) presents the system parameters used in the experiments, which are based on real-world data.

6.2. Performance of the LBB algorithm and the ALNS&TS heuristic

In this section, we conduct numerical experiments on instance sets IS1, IS2, and IS3. By comparing the solutions obtained using the proposed methods with those generated by Gurobi, we validate the effectiveness of both the LBB algorithm and the ALNS&TS heuristic.

[Table 5](#) presents the results, where $Obj_Gap = (Best_Obj - Obj(Gurobi)) / Best_Obj$. The results demonstrate that the LBB algorithm outperforms Gurobi in terms of computational efficiency and the scale of solvable problems. Specifically, the LBB algorithm is, on average, 3.31 times faster than Gurobi when solving instance sets IS1 and IS2. Additionally, the LBB algorithm is capable of solving larger-scale problems to optimality. For instance, Gurobi fails to generate any solutions for instance set IS3 within 30 min, whereas the LBB algorithm provides optimal solutions for this set. [Table 5](#) further validates the efficiency of the ALNS&TS heuristic. Across the three instance sets, the ALNS&TS heuristic delivers high-quality solutions in an average computation time of just 1.17 s. The average relative error between the ALNS&TS heuristic and the optimal solution is a mere 1.94%, confirming the heuristic's effectiveness in balancing speed and solution quality.

6.3. Performance of the ALNS&TS heuristic on large-scale instances

To evaluate the effectiveness of the proposed ALNS&TS heuristic on large-scale instances, a benchmark comparison is conducted against a tailored genetic algorithm (GA). In this GA, the task sequences of shuttles and shuttle-lifts are independently encoded as two distinct chromosomes. The task sequence of the bin-lift is implicitly determined during the decoding phase based on the arrival order of bins at the bin-lift buffer, consistent with the assignment rule adopted in the ALNS&TS approach. Genetic operators such as crossover and mutation are applied separately to the two chromosomes. The integrated decoding procedure reconstructs the complete schedule and evaluates its feasibility and performance (e.g., makespan). Infeasible solutions are penalized during fitness evaluation to guide the evolutionary search toward high-quality schedules.

[Table 6](#) provides a detailed performance comparison between the ALNS&TS heuristic and the GA across various large-scale instances. The results indicate a marked improvement in operational efficiency achieved by the ALNS&TS heuristic. Specifically, the makespan produced by the ALNS&TS heuristic is, on average, 55.13% shorter than that of the GA. This significant reduction underscores the effectiveness of the ALNS&TS heuristic in optimizing task scheduling and resource allocation, particularly in complex and large-scale scenarios where traditional methods like GA struggle to perform efficiently. The instance sizes evaluated in this study, with a maximum of 110 retrieval requests, align well with the scheduling requirements for a single wave of orders. Wave picking ([Rasmi et al., 2022](#)), a commonly employed operational strategy, involves dividing daily orders into multiple waves and processing each wave collectively. This picking approach not only improves picking efficiency but also streamlines subsequent operations such as packing, sorting, and shipping.

6.4. Analyzing the effect of the number configuration of shuttles and lifts on the makespan

To first examine the impact of the number of shuttles and lifts (shuttle-lifts and bin-lifts) on the system makespan, a sensitivity analysis is conducted under three levels of retrieval requests ($N_R \in \{20, 30, 40\}$). For each case, three values are assigned to both shuttle-lifts and bin-lifts, while the number of shuttles is varied across four levels (e.g., $N_S \in \{2, 3, 4, 5\}$ when $N_{L_S} = 3$). First, given a fixed number of shuttles and shuttle-lifts, the number of bin-lifts is incremented from 2 to N_S in steps of one to observe changes in the system makespan. Similarly, with the number of shuttles and bin-lifts fixed, the number

Table 3
Description of instance sets.

Instance set	Number of requests	Number of shuttles	Number of shuttle-lifts	Number of bin-lifts
IS1	5	2	2	2
IS2	7	2	2	2
IS3	10	3	2	2
IS4	20	4	3	3
IS5	40	6	4	4
IS6	60	8	5	5
IS7	80	10	6	6
IS8	100	12	7	7
IS9	110	13	8	8

Table 4
System parameters used in the experiments.

H	W	L	he	wi	le	v_S	v_{L_S}	v_{L_B}	t_{lu}
25 (bins)	40 (bins)	100 (bins)	0.33 (m)	0.61 (m)	0.81 (m)	3 (m/s)	3 (m/s)	5 (m/s)	7 (s)

Table 5
Performance of the LBBD algorithm and the ALNS&TS heuristic.

Instance	Index	Gurobi		LBBD		ALNS&TS Heuristic			
		Obj (s)	Time (s)	Obj (s)	Time (s)	Initial_Obj (s)	Best_Obj (s)	Obj_Gap (%)	Time (s)
IS1	1	158.55	16.47	158.55	0.17	176.70	158.55	0.00	0.50
	2	165.14	9.42	165.14	2.63	221.52	165.14	0.00	0.43
	3	150.52	20.70	150.52	2.48	218.09	152.20	1.11	0.31
	4	134.96	13.55	134.96	4.50	194.14	134.96	0.00	0.27
	5	138.26	5.26	138.26	1.81	223.84	138.58	0.23	0.77
IS2	1	175.10	420.36	175.10	170.60	265.58	183.12	4.38	0.78
	2	226.82	1332.68	226.82	238.99	407.85	236.00	3.89	1.05
	3	213.64	976.09	213.64	401.54	331.74	213.64	0.00	0.41
	4	212.81	449.43	212.81	233.09	269.77	217.82	2.30	0.36
	5	197.48	599.75	197.48	105.02	304.28	204.39	3.38	0.36
IS3	1	–	1800.00	249.25	1074.38	424.31	257.09	3.05	3.27
	2	–	1800.00	216.99	1006.56	344.86	228.35	4.97	2.43
	3	–	1800.00	233.37	1073.54	331.68	241.77	3.48	2.11
	4	–	1800.00	219.92	1040.02	393.71	223.98	1.81	2.10
	5	–	1800.00	221.21	1025.62	320.35	222.36	0.52	2.35
Average		–	826.76	194.27	406.06	295.23	198.53	1.94	1.17

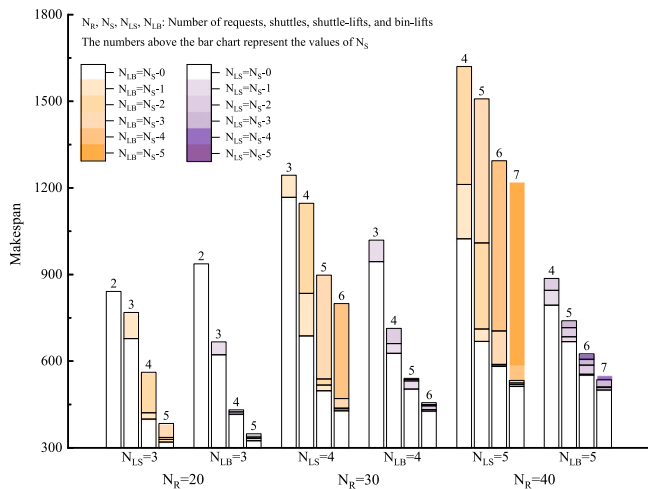


Fig. 6. Analyzing the Effect of the Number of Shuttles and Lifts on the Makespan.

of shuttle-lifts is increased incrementally from 2 to N_S , and the corresponding changes in makespan are recorded. The results of this analysis are presented in Fig. 6. The orange-shaded bars indicate reductions in makespan due to increases in the number of bin-lifts, while the purple-shaded bars represent reductions resulting from increases in shuttle-lifts.

As shown in Fig. 6, the system makespan significantly decreases as the number of shuttles increases, although the marginal effect diminishes with each additional shuttle. The reduction in makespan achieved by increasing the number of bin-lifts is comparable to that achieved by adding more shuttles, and both reductions are greater than those resulting from an increase in shuttle-lifts. Thus, when system investment is limited but minimizing the makespan is a priority, it is recommended to prioritize increasing the number of shuttles or bin-lifts. Additional shuttle-lifts can be considered as a secondary option.

6.5. Analyzing the effect of investment allocation between bin-lifts and shuttle-lifts on the makespan

In practice, decision-makers are often more concerned with how capital investment translates into operational efficiency. Given the notable disparity in unit prices between shuttle-lifts and bin-lifts, it becomes important to further examine the relationship between equipment investment cost and makespan. To complement the quantity-based analysis and provide more practical decision-making support, we design two additional experiments focused on the cost-makespan trade-off. In the first experiment, we examine the effect of incremental investment in each type of lift while keeping the number of the other type constant. Specifically, we increase the investment in shuttle-lifts and bin-lifts separately by RMB 150,000, 300,000, 450,000, 600,000, and 750,000, and observe the corresponding changes in makespan. This setting maintains consistent baseline equipment levels and allows us to isolate the marginal impact of each lift type when additional capital is allocated. In the second experiment, we consider the scenario where

Table 6
Performance of ALNS&TS heuristic on large-scale instances.

Instance	Index	GA		ALNS&TS		
		Obj (s)	Time (s)	Obj (s)	Time (s)	Reduction in Obj (%)
IS4	1	381.11	28.62	359.20	2.57	6.10
	2	424.84	26.38	377.14	3.34	12.65
	3	420.70	26.53	387.51	2.89	8.56
	4	396.85	26.22	358.58	3.68	10.68
	5	439.55	27.29	405.38	3.30	8.43
	Average	412.61	27.01	377.56	3.15	9.28
	Standard Deviation	20.89	0.89	17.72	0.39	2.22
IS5	1	686.53	87.24	554.46	30.16	23.82
	2	777.90	86.05	531.59	25.37	46.33
	3	717.07	85.46	558.21	26.85	28.46
	4	761.87	87.10	531.67	27.85	43.30
	5	730.24	86.56	518.26	26.70	40.90
	Average	734.72	86.48	538.84	27.38	36.56
	Standard Deviation	323.74	59.48	161.99	24.28	28.66
IS6	1	1043.82	204.35	658.01	113.49	58.63
	2	1052.35	202.34	668.44	101.03	57.43
	3	1003.32	203.75	604.61	111.35	65.95
	4	1033.67	206.05	630.80	107.73	63.87
	5	971.17	195.58	672.70	101.60	44.37
	Average	1020.86	202.41	646.91	107.04	58.05
	Standard Deviation	608.99	175.44	270.58	104.01	49.34
IS7	1	1170.36	407.59	743.28	286.85	57.46
	2	1106.44	406.77	695.96	275.52	58.98
	3	1226.25	407.46	719.18	258.14	70.51
	4	1194.04	410.13	639.93	306.24	86.59
	5	1291.77	410.93	724.80	257.71	78.22
	Average	1197.77	408.58	704.63	276.89	70.35
	Standard Deviation	787.55	381.57	329.01	274.35	62.08
IS8	1	1551.94	760.75	784.58	701.93	97.80
	2	1523.75	763.12	759.43	637.83	100.64
	3	1314.77	771.83	799.61	553.99	64.43
	4	1433.44	781.17	787.50	723.84	82.02
	5	1445.99	777.71	859.11	653.37	68.31
	Average	1453.98	770.92	798.05	654.19	82.64
	Standard Deviation	1044.66	743.95	421.80	653.71	74.83
IS9	1	1307.56	1074.83	762.77	963.77	71.42
	2	1407.10	1068.01	758.48	974.22	85.52
	3	1590.22	1057.32	879.18	921.57	80.88
	4	1518.13	1037.33	830.53	936.67	82.79
	5	1431.38	1047.24	778.78	926.35	83.80
	Average	1450.88	1056.94	801.95	944.51	80.88
	Standard Deviation	1042.76	1030.03	426.91	941.59	71.77

total initial investment in shuttle-lifts and bin-lifts is equal, taking into account their differing unit costs. Because bin-lifts are less expensive than shuttle-lifts, a given amount of capital allows for the addition of more bin-lifts than shuttle-lifts. This experiment ensures a fair comparison by matching total cost rather than quantity, thus better reflecting real-world investment decision-making where budget constraints are often a key consideration. The equipment costs are obtained from real-world cases, with the investment costs specified as follows: (i) Shuttle: 90,000 RMB; (ii) Shuttle-lift: 190,000 RMB; (iii) Bin-lift: 120,000 RMB. These costs are annualized over 7 years with an assumed interest rate of $R = 0.5\%$. The annual costs for each type of equipment are calculated using the formula: $C_S = \sum_{i=1}^7 (90000(1+R)^{i-1})/7$, $C_{LS} = \sum_{i=1}^7 (190000(1+R)^{i-1})/7$, $C_{LB} = \sum_{i=1}^7 (120000(1+R)^{i-1})/7$.

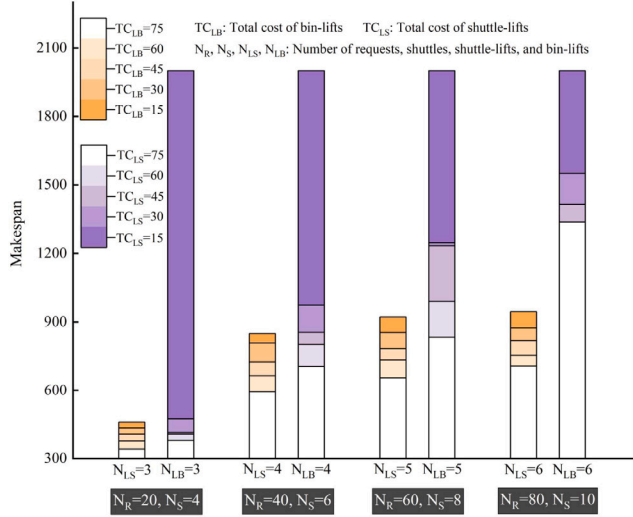
Experimental results in Fig. 7 consistently reveal that investment in bin-lifts yields greater performance improvement per unit cost. In the first experiment, where the initial number of shuttle-lifts and bin-lifts is kept equal, increasing investment in bin-lifts leads to a more significant reduction in makespan compared to shuttle-lifts. This suggests that, under a quantity-equal setting, bin-lifts offer a higher marginal return on investment. In the second experiment, where the initial investment cost of the two types of lifts is equal rather than their quantity, the performance advantage of bin-lifts is still observed, though the extent of makespan reduction becomes less pronounced. This weakening is due to the lower unit cost of bin-lifts: under the same budget, adding

shuttle-lifts results in a smaller number of new units but allows for a larger number of bin-lifts to be configured in parallel. Nevertheless, across both experimental settings, bin-lifts consistently demonstrate a superior cost-performance ratio, reinforcing the recommendation to prioritize bin-lift investment when aiming to reduce makespan under constrained budgets.

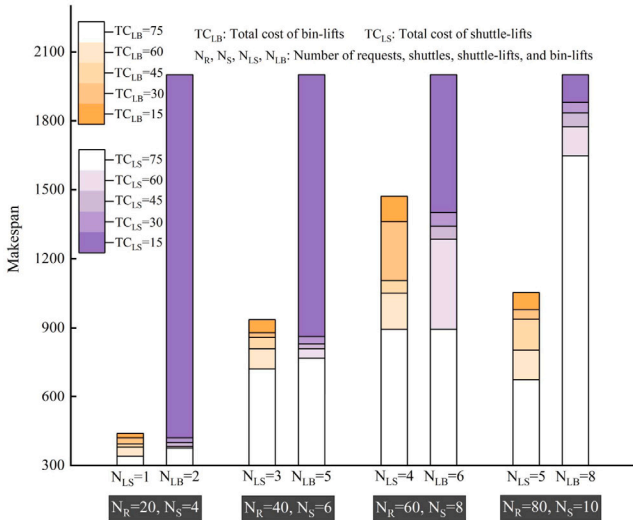
To investigate the optimal allocation of a fixed investment budget between different types of lifts, we further evaluate system makespan under varying combinations of shuttle-lifts and bin-lifts. In this analysis, the total investment amount is held constant, while the proportion of cost allocated to bin-lifts (referred to as *cost ratio*) is varied across five levels: 0, 0.2, 0.4, 0.6, and 0.8. For each allocation scheme, the quantities of shuttle-lifts and bin-lifts are adjusted accordingly based on their respective unit costs. The objective is to assess whether partially substituting shuttle-lifts with bin-lifts under budget constraints can enhance overall system performance, and to identify the impact of different cost ratios on system makespan. Fig. 8 presents the resulting makespan under each cost ratio, considering various configurations of N_R and N_S . The results indicate that resource composition under a fixed cost budget has a substantial impact on operational efficiency. Notably, partial substitution of shuttle-lifts with bin-lifts reduces makespan in most scenarios. Among the tested configurations, allocating 20% of the total investment to bin-lifts (i.e., cost ratio = 0.2) yields the greatest improvement in performance, suggesting an optimal balance between the

Table 7
Cost-benefit summary of lift investment strategies.

Experiment	Investment strategy description	Metric	Key finding
1	Equal initial number.	Equal initial number.	Investment in bin-lifts leads to greater makespan reduction than shuttle-lifts.
2	Equal initial cost.	Equal initial cost.	Allocate 20% of investment to bin-lifts achieves optimal makespan reduction.
3	Fixed total budget, vary cost ratio between bin-lifts and shuttle-lifts.	Fixed total budget.	



(a) Evaluating Cost Efficiency under Equal Initial Lift Quantities



(b) Evaluating Cost Efficiency under Equal Initial Lift Costs

Fig. 7. Cost-Effectiveness of Lift Investments: Bin-Lifts vs. Shuttle-Lifts.

two lift types. This implies that a moderate deployment of bin-lifts can effectively alleviate system bottlenecks while preserving the throughput advantages provided by shuttle-lifts. However, further increasing the bin-lift proportion beyond this point leads to a deterioration in system performance, primarily due to the shortage of shuttle-lifts required for efficient material handling.

Table 7 consolidates evidence from three experiments and yields two complementary managerial rules. First, on a marginal basis (whether measured per added unit or per unit cost), investments in bin-lifts produce larger makespan reductions than investments in shuttle-lifts. When capital is tight, prioritize bin-lifts to relieve bin tier-switching queues. Second, for a one-shot fixed total budget, the

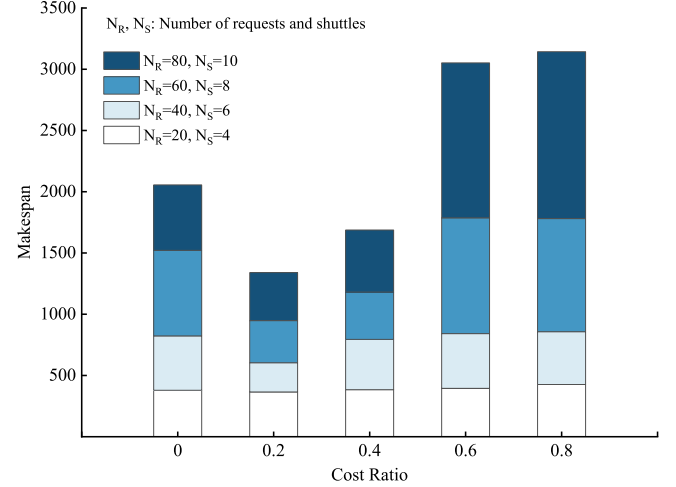


Fig. 8. Effect of Bin-Lift Cost Ratio on System Performance with a Fixed Budget.

near-optimal share for bin-lifts is about 20%. Over-allocating bin-lifts can shift the bottleneck to shuttle cross-tier switching. Once bin tier-switching delay no longer dominates, deploy additional funds to shuttle-lifts to cut cross-tier switching time and rebalance tier workloads.

6.6. Analyzing the effect of different configurations of lift location on the makespan

In the system, the locations of the two types of lifts significantly influence the system makespan. Specifically, the locations of the bin-lifts correspond to the I/O points, making the configuration of these points a critical issue in automated warehouses (Jen et al., 2024; Xu et al., 2020). While the locations of shuttle-lifts do not directly affect the bin execution process, they impact the waiting time when a shuttle needs to switch operating tiers. Therefore, optimizing the locations of both types of lifts is essential for enhancing the system's operational efficiency. In practice, managers often prefer placing lifts in the middle of the system rather than at the ends of the picking tracks, as this approach can reduce travel distances. Beyond the conventional configuration (referred to as Configuration 1), where lifts are located at opposite ends of the picking tracks, this section introduces four alternative lift configurations. This analysis aims to evaluate the impact of these configurations on the system makespan and provide insights for optimizing lift placement.

(i) Configuration 2: Both types of lifts are located at the same end of the picking tracks.

(ii) Configuration 3: Shuttle-lifts are uniformly distributed at the system's midpoint along the X-axis, while bin-lifts remain at the ends of the picking tracks.

(iii) Configuration 4: Bin-lifts are uniformly distributed at the system's midpoint along the X-axis, while shuttle-lifts are positioned at the ends of the picking tracks.

(iv) Configuration 5: Both shuttle-lifts and bin-lifts are uniformly distributed at the system's midpoint along the X-axis.

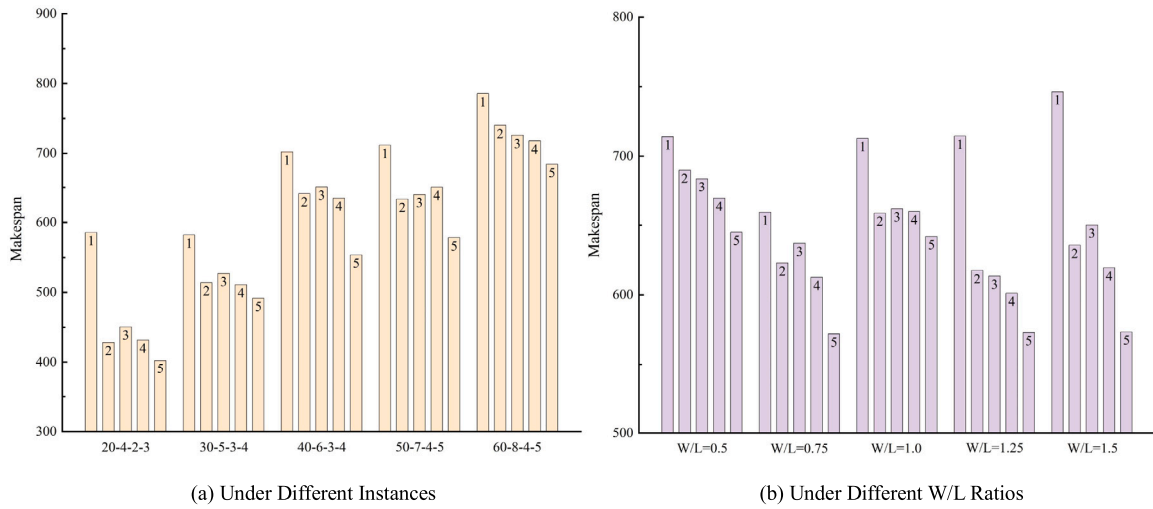


Fig. 9. Comparison of the Makespan under Five Configurations of Lift Location.

This section examines the impact of different lift location configurations on the system makespan under varying instances (Fig. 9(a)) and different width/length (W/L) ratios (Fig. 9(b)). The bar charts in Fig. 9 represent the system makespan for each lift location configuration, with the bars from left to right corresponding to Configurations 1 through 5. As shown in Fig. 9, Configuration 5 consistently achieves the shortest makespan across all instances. Configurations 2–4 exhibit comparable performances, all of which outperform Configuration 1. Notably, as the W/L ratio increases, placing the two types of lifts at opposite ends of the picking tracks (Configuration 1) becomes less effective. This arrangement forces shuttles to traverse the entire system along the X-axis when changing operating tiers, leading to a prolonged makespan.

6.7. Performance comparison of the ALNS&TS heuristic and other common scheduling strategies

To validate the effectiveness of the ALNS&TS heuristic, we compare its performance with three commonly used scheduling strategies: the increasing travel time (ITT) strategy, the first-come-first-served (FCFS) strategy, and the retrieval sequence consistency (RSC) strategy.

- ITT strategy: Tasks are prioritized based on their required operation time, with tasks requiring shorter operation times being executed first.
- FCFS strategy: Tasks are executed in the order they are generated. This practical strategy prioritizes retrieval requests based on their arrival time for assignment to shuttles. The sequence of shuttle arrivals at the shuttle-lift buffer determines the order of shuttle-transfer tasks, while the arrival order of bins at the bin-lift buffer dictates the execution sequence of vertical-retrieval tasks.
- RSC strategy: The task sequence for shuttles is randomly generated initially. Both shuttle-lifts and bin-lifts are then required to execute tasks in the same order as determined by the shuttle's task sequence.

In addition to comparing the makespan of the ALNS&TS heuristic with the three other strategies, we also evaluate their average retrieval time. Fig. 10 illustrates the performance of these strategies in terms of makespan and average retrieval time. The bar charts represent the makespan (orange-shaded bars) and average retrieval time (purple-shaded bars) for each strategy, while the line charts indicate the percentage reduction in makespan and average retrieval time achieved by the ALNS&TS heuristic compared to the other strategies. As shown in Fig. 10, the ALNS&TS heuristic significantly outperforms the ITT strategy in both metrics, achieving substantial reductions. While the

FCFS and RSC strategies perform better than the ITT strategy, they still fall short when compared to the ALNS&TS heuristic. On average, the makespan of the FCFS and RSC strategies is approximately 16.25% and 18.72% higher than that of the ALNS&TS heuristic, respectively. Similarly, the average retrieval time for these two strategies is about 68.49% and 78.35% higher, respectively, highlighting the superior efficiency of the ALNS&TS heuristic.

To evaluate the cost-efficiency of the four strategies, we further analyze the minimum required investment cost to meet the makespan requirements. Since the strategies vary only in the number of equipment needed, we focus on the equipment investment costs. Fig. 11 illustrates the investment costs associated with the ALNS&TS heuristic and three other strategies. The bar chart represents the minimum investment cost, while the line chart shows the percentage reduction in investment cost achieved by the ALNS&TS heuristic relative to the other strategies. For example, in instance 20-4-3-3, all strategies must meet the constraint that the maximum completion time does not exceed 400 s while minimizing investment cost. As shown in Fig. 11, the ALNS&TS heuristic delivers an average investment cost reduction of 36.08% compared to the three alternative strategies. Specifically, the ALNS&TS heuristic achieves the most significant cost savings when compared to the ITT strategy, with an average cost reduction ratio of 88.50%. The investment costs for the FCFS and RSC strategies are similar, being 8.51% and 11.24% higher, respectively, than the investment cost of the ALNS&TS heuristic. This highlights the superior cost-efficiency of the ALNS&TS heuristic in meeting scheduling constraints while minimizing costs.

7. Conclusion

Prior scheduling literature on AS/RSs has largely focused on single-resource settings or weakly coupled two-resource cases, often abstracting away cross-resource timing. In contrast, the F-SBS/RS-TL requires tight synchronization among three distinct resources: four-directional shuttles, shuttle-lifts, and bin-lifts. Inter-tier precedence makes the makespan depend on system-wide temporal coordination rather than the capacity of any single resource. By explicitly modeling and optimizing this coupling, we address retrieval scheduling in the F-SBS/RS-TL. The goal is to sequence retrieval requests and jointly schedule shuttles, shuttle-lifts, and bin-lifts to minimize the makespan. We formulate an MIP model and design an exact LBBP to solve it. Specifically, the Benders master problem determines the optimal execution order of retrieval requests and the task sequence of shuttles. Based on these solutions, the Benders subproblem identifies the optimal task sequence

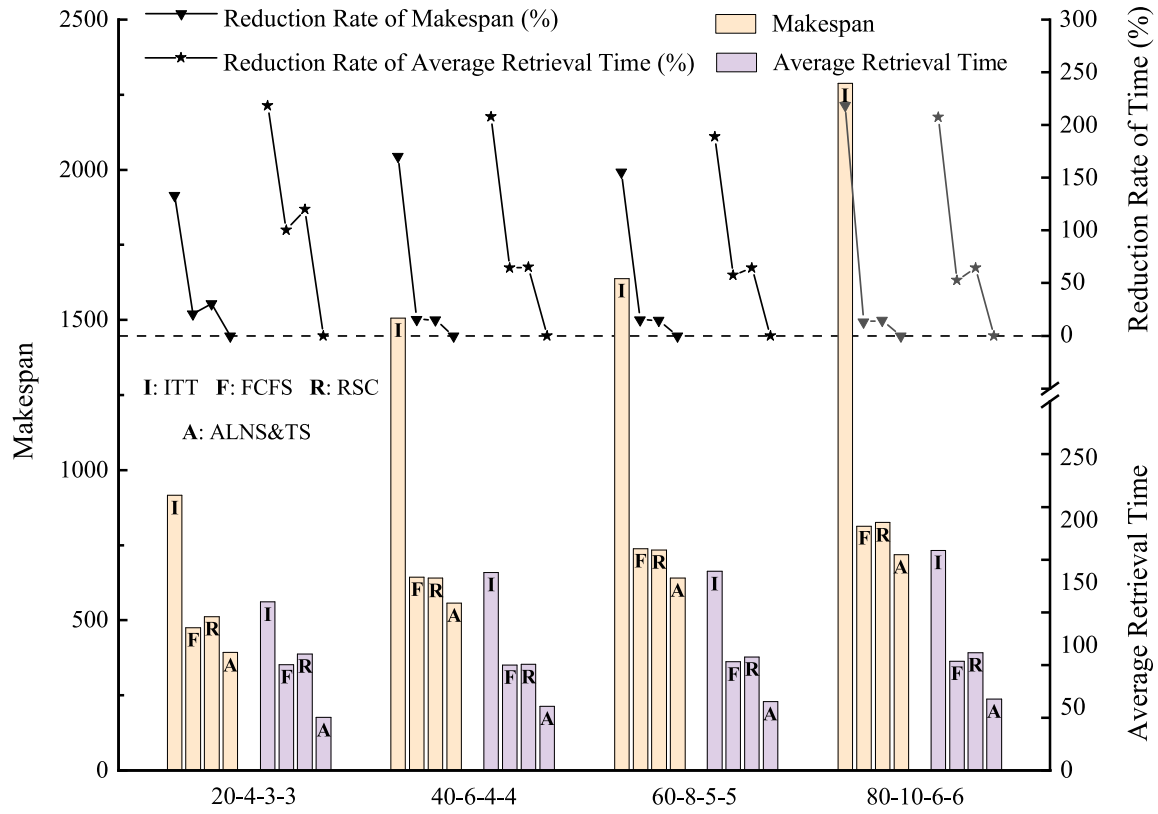


Fig. 10. Efficiency Comparison of the ALNS&TS Heuristic and Other Common Strategies.

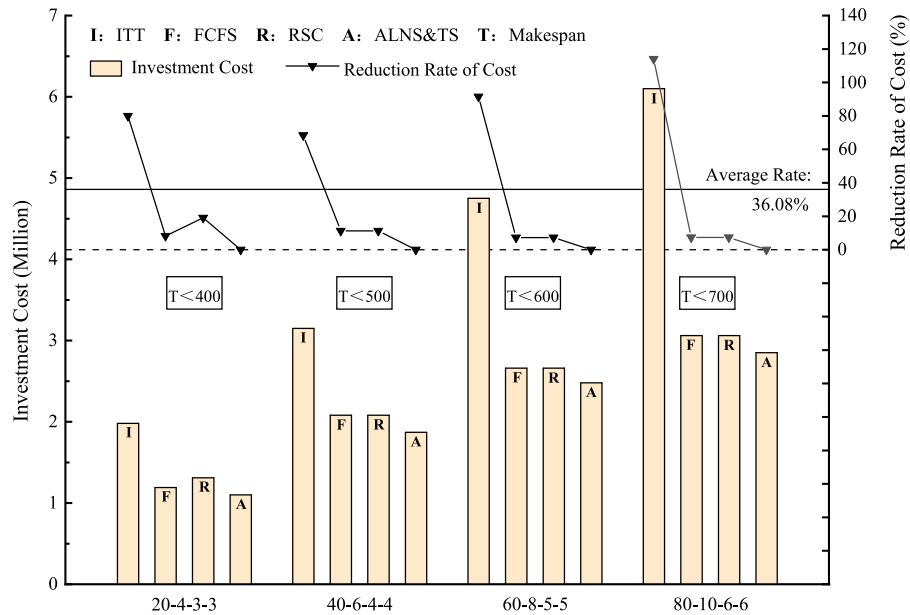


Fig. 11. Cost Comparison of the ALNS&TS Heuristic and Other Common Strategies.

for lifts. Numerical experiments demonstrate that the LBB algorithm is 3.31 times faster than the commercial solver. Additionally, we design a two-stage heuristic, ALNS&TS, to handle large-scale instances. This heuristic achieves high-quality solutions within a short computational time, reducing the makespan by an average of 55.13% compared to the GA.

Our findings provide valuable insights for warehouse designers and managers of the F-SBS/RS-TL. (i) *Equipment Quantity Configuration*: Increasing the number of shuttles, shuttle-lifts, and bin-lifts initially

reduces the completion time for all retrieval requests. However, after reaching a threshold, further increases result in diminishing marginal improvements to the system makespan. Notably, increasing the number of shuttles and bin-lifts yields a more significant reduction in makespan compared to adding shuttle-lifts. (ii) *Equipment Cost-Effectiveness Analysis*: Partial substitution of shuttle-lifts with bin-lifts under a fixed investment budget yields notable improvements in system efficiency. The most pronounced benefits are achieved when bin-lifts constitute 20% of the total lift-related investment, indicating an optimal balance

between cost and performance. (iii) *Lift Location Configuration*: Placing both types of lifts (shuttle-lifts and bin-lifts) at the midpoint of the system along the X-axis consistently minimizes the makespan across all tested instances. (iv) *Efficient Scheduling Strategy*: The ALNS&TS heuristic outperforms common scheduling strategies in terms of efficiency and cost. Compared to the ITT strategy, the ALNS&TS heuristic significantly reduces system makespan, average retrieval time, and investment cost. Relative to the FCFS and RSC strategies, the heuristic reduces the makespan by 16.25% and 18.72%, the average retrieval time by 68.49% and 78.35%, and the investment cost by 8.51% and 11.24%, respectively.

Despite the strengths of this study, several limitations should be acknowledged as potential directions for future work. (i) Blocking effects among four-directional shuttles are not considered, which may affect system performance under compact layouts. (ii) The model assumes a static, offline setting with known retrieval sequences and excludes storage tasks and shuttle rebalancing, limiting its applicability in dynamic environments. (iii) The scheduling problem is studied in isolation, whereas in practice, it closely interacts with other warehousing operations such as storage assignment and routing. These limitations point to several promising directions for future research, as discussed below.

(i) Mitigating Blocking Effects: In the F-SBS/RS-TL, blocking among shuttles remains a critical issue affecting system throughput. Future work could develop advanced scheduling strategies or coordination protocols to mitigate shuttle blocking and incorporate dynamic rebalancing to improve efficiency under asymmetric workloads. Incorporating predictive models or decentralized control strategies may also help achieve more seamless operations.

(ii) Dynamic and Real-Time Scheduling: Current scheduling strategies are often static or offline. Future studies may investigate dynamic, real-time scheduling algorithms that respond adaptively to order characteristics and arrival patterns. This may involve using rolling horizon optimization, reinforcement learning, or online optimization approaches to handle system uncertainty and operational variability.

(iii) Integrated System-Level Optimization: To further improve the performance of F-SBS/RS-TL systems, future research could focus on integrated design and optimization approaches that jointly consider storage assignment, routing, and scheduling strategies. The use of data-driven simulation–optimization frameworks, queueing models, or digital twin technologies could provide valuable insights for system design and operation.

CRediT authorship contribution statement

Rong Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Peiran Tao**: Validation, Software, Methodology, Investigation. **Rui Chen**: Writing – review & editing, Supervision, Resources, Formal analysis, Data curation. **Taoxing Zhu**: Visualization, Resources, Data curation. **Peng Yang**: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

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Data availability

Data will be made available on request.

References

- Arslan, B., & Ekren, B. Y. (2023). Transaction selection policy in tier-to-tier SBS/RS by using deep Q-learning. *International Journal of Production Research*, 61(21), 7353–7366.
- Benders, J. F. (1962). Partitioning procedures for solving mixedvariables programming problems. *Numerische Mathematik*, 4, 238–252.
- Boysen, N., Briskorn, D., & Emde, S. (2017). Sequencing of picking orders in mobile rack warehouses. *European Journal of Operational Research*, 259(1), 293–307.
- Boysen, N., De Koster, R., & Weidinger, F. (2019). Warehousing in the e-commerce era: A survey. *European Journal of Operational Research*, 277(2), 396–411.
- Boysen, N., & Stephan, K. (2016). A survey on single crane scheduling in automated storage/retrieval systems. *European Journal of Operational Research*, 254(3), 691–704.
- Carlo, H. J., & Vis, I. F. (2012). Sequencing dynamic storage systems with multiple lifts and shuttles. *International Journal of Production Economics*, 140(2), 844–853.
- Chen, R., Yang, J., Yu, Y., & Guo, X. (2023). Retrieval request scheduling in a shuttle-based storage and retrieval system with two lifts. *Transportation Research Part E*, 174, Article 103117.
- De Koster, R., Le-Duc, T., & Yu, Y. (2008). Optimal storage rack design for a 3-dimensional compact AS/rs. *International Journal of Production Research*, 46(6), 1495–1514.
- Dong, W., Jin, M., Wang, Y., & Kelle, P. (2021). Retrieval scheduling in crane-based 3D automated retrieval and storage systems with shuttles. *Annals of Operations Research*, 193(1), 111–135.
- Ekren, B. Y., & Arslan, B. (2024). A reinforcement learning approach for transaction scheduling in a shuttle-based storage and retrieval system. *International Transactions in Operational Research*, 31, 274–295.
- Ekren, B. Y., Lerher, T., Küçükyavaş, M., & Jerman, B. (2024). Cost and performance comparison of tier-captive SBS/RS with a novel AVS/RS/ml. *International Journal of Production Research*, 62(5), 1648–1662.
- Enfon (2025). Intelligent multishuttle system. <http://www.enfon.com.cn/pszx>.
- Francois, V., Arda, Y., & Crama, Y. (2019). Adaptive large neighborhood search for multitrip vehicle routing with time windows. *Transportation Science*, 53(6), 1706–1730.
- Gendreau, M., & Potvin, J. Y. (2010). Tabu search. In *Handbook of metaheuristics* (pp. 41–59). Boston, MA: Springer US.
- Gharehgozli, A. H., Yu, Y., Zhang, X., & De Koster, R. (2017). Polynomial time algorithms to minimize total travel time in a two-depot automated storage/retrieval system. *Transportation Science*, 51(1), 19–33.
- Glover, F. (1986). Future paths for integer programming and links to artificial intelligence. *Computers & Operations Research*, 13(5), 533–549.
- Gurobi Optimizer (2025). Gurobi optimizer: The world's fastest solver. <https://www.gurobi.com/solutions/gurobi-optimizer/>.
- Han, M. H., McGinnis, L. F., Shieh, J. S., & White, J. A. (1987). On sequencing retrievals in an automated storage/retrieval system. *IIE Transactions*, 19(1), 56–66.
- Hoogeveen, J. A., Lenstra, J. K., & Veltman, B. (1996). Preemptive scheduling in a two-stage multiprocessor flow shop is NP-hard. *European Journal of Operational Research*, 89(1), 172–175.
- Hooker, J. N. (2000). *Logic-based methods for optimization: Combining optimization and constraint satisfaction*. New York: Wiley.
- Hooker, J. N., & Ottosson, G. (2003). Logic-based benders decomposition. *Mathematical Programming*, 96(1), 33–60.
- Jen, T. K., Wang, R., & Yang, P. (2024). Determining the I/O point policy in a robotic live-cube compact storage system. *International Journal of Production Research*.
- Jerman, B., Ekren, B. Y., Küçükyavaş, M., & Lerher, T. (2021). Simulation-based performance analysis for a novel AVS/RS technology with movable lifts. *Applied Sciences*, 11(5), 1–16.
- Kosanić, N., Marolt, J., Zrnić, N., & Lerher, T. (2023). Travel time model for multiple-deep shuttle-based storage and retrieval systems. *International Journal of Production Research*, 62(7), 2606–2639.
- Lerher, T. (2015). Travel time model for double-deep shuttle-based storage and retrieval systems. *International Journal of Production Research*, 54(9), 2519–2540.
- Lerher, T., Ficko, M., & Palčić, I. (2021). Throughput performance analysis of automated vehicle storage and retrieval systems with multiple-tier shuttle vehicles. *Applied Mathematical Modelling*, 91, 1004–1022.
- Li, X., & Gao, L. (2016). An effective hybrid genetic algorithm and tabu search for flexible job shop scheduling problem. *International Journal of Production Economics*, 174, 93–110.
- Man, X., Zheng, F., Chu, F., Liu, M., & Xu, Y. (2021). Bi-objective optimization for a two-depot automated storage/retrieval system in container terminals. *Annals of Operations Research*, 296, 243–262.

- Marolt, J., Šinko, S., & Lerher, T. (2023). Model of a multiple-deep automated vehicles storage and retrieval system following the combination of depth-first storage and depth-first relocation strategies. *International Journal of Production Research*, 61(15), 4991–5008.
- Nia, A. R., Haleh, H., & Saghaei, A. (2017). Dual command cycle dynamic sequencing method to consider GHG efficiency in unit-load multiple-rack automated storage and retrieval systems. *Computers & Industrial Engineering*, 111, 89–108.
- Polten, L., & Emde, S. (2022). Multi-shuttle crane scheduling in automated storage and retrieval systems. *European Journal of Operational Research*, 302(3), 892–908.
- Rasmi, S. A. B., Wang, Y., & Charkhgard, H. (2022). Wave order picking under the mixed-shelves storage strategy: A solution method and advantages. *Computers & Operations Research*, 137, Article 105556.
- Roodbergen, K. J., & Vis, I. F. A. (2009). A survey of literature on automated storage and retrieval systems. *European Journal of Operational Research*, 194(2), 343–362.
- Ropke, S., & Pisinger, D. (2006). An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. *Transportation Science*, 40(4), 455–472.
- Ropke, S., & Pisinger, D. (2006). A unified heuristic for a large class of vehicle routing problems with backhauls. *European Journal of Operational Research*, 171(3), 750–775.
- Roshanaei, V., Booth, K. E. C., Aleman, D. M., Urbach, D. R., & Beck, J. C. (2020). Branch-and-check methods for multi-level operating room planning and scheduling. *International Journal of Production Economics*, 220, Article 107433.
- Shaw, P. (1997). *A new local search algorithm providing high quality solutions to vehicle routing problems: vol. 46*. Glasgow, Scotland, UK: APES Group, Dept of Computer Science, University of Strathclyde.
- Soriano, A., Gansterer, M., & F., H. R. (2023). The multi-depot vehicle routing problem with profit fairness. *International Journal of Production Economics*, 255, Article 108669.
- Tappia, E., Roy, D., De Koster, R., & Melacini, M. (2016). Modeling, analysis, and design insights for shuttle-based compact storage systems. *Transportation Science*, 51(1), 269–295.
- Vis, I. F. A., & Roodbergen, K. J. (2009). Scheduling of container storage and retrieval. *Operations Research*, 57(2), 456–467.
- Wang, Y., Liu, Z., Huang, K., Mou, S., & Zhang, R. (2020). Model and solution approaches for retrieval operations in a multi-tier shuttle warehouse system. *Computers & Industrial Engineering*, 141, Article 106283.
- Wayzim (2025). *Intelligent warehousing*. <https://wayzim.com/en/solution/detail?id=128>.
- Xu, X., Zhao, X., Zou, B., Gong, Y., & Wang, H. (2020). Travel time models for a three-dimensional compact AS/RS considering different I/O point policies. *International Journal of Production Research*, 58(18), 5432–5455.
- Yang, J., de Koster, R., Guo, X., & Yu, Y. (2023). Scheduling shuttles in deep-lane shuttle-based storage systems. *European Journal of Operational Research*, 308(2), 696–708.
- Yu, Y., Yang, J., & Guo, X. (2023). Retrieval sequencing in autonomous vehicle storage and retrieval systems. *International Journal of Production Research*, 1–20.
- Zaerpour, N., Yu, Y., & De Koster, R. (2015). Storing fresh produce for fast retrieval in an automated compact cross-dock system. *Production and Operations Management*, 24(8), 1266–1284.
- Zhao, N., Luo, L., & Lodewijks, G. (2018). Scheduling two lifts on a common rail considering acceleration and deceleration in a shuttle based storage and retrieval system. *Computers & Industrial Engineering*, 124, 48–57.
- Zhen, L., Wu, J., Li, H., Tan, Z., & Yuan, Y. (2023). Scheduling multiple types of equipment in an automated warehouse. *Annals of Operations Research*, 322, 1119–1141.
- Zou, B., De Koster, R., & Xu, X. (2018). Operating policies in robotic compact storage and retrieval systems. *Transportation Science*, 52(4), 788–811.
- Zou, B., Xu, X., Gong, Y., & De Koster, R. (2016). Modeling parallel movement of lifts and vehicles in tier-captive vehicle-based warehousing systems. *European Journal of Operational Research*, 254(1), 51–67.